

ROBOSIM

Project team: Freya, Cassie, Raymond, Mincheol, Ziyang

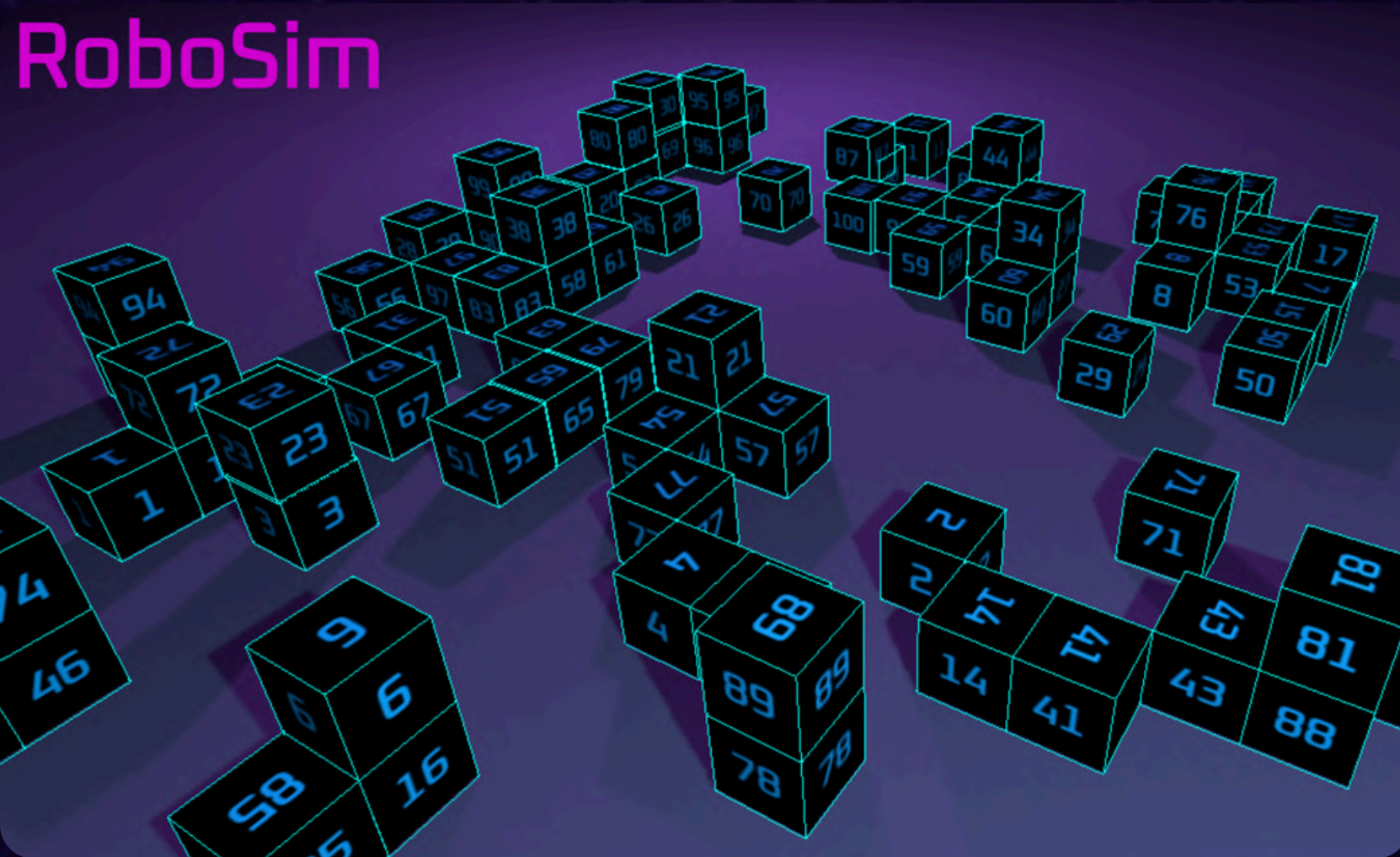
OUR CLIENT



- Marius Jurt: Master's Student at Bristol, Research Engineer at Toshiba when Robosim was started
- Project previously run by Toshiba, Marius continued it out of passion when Toshiba dropped it
- **Fourth year running!**
- Gave us a lot of flexibility in development

WHAT IS ROBOSIM?

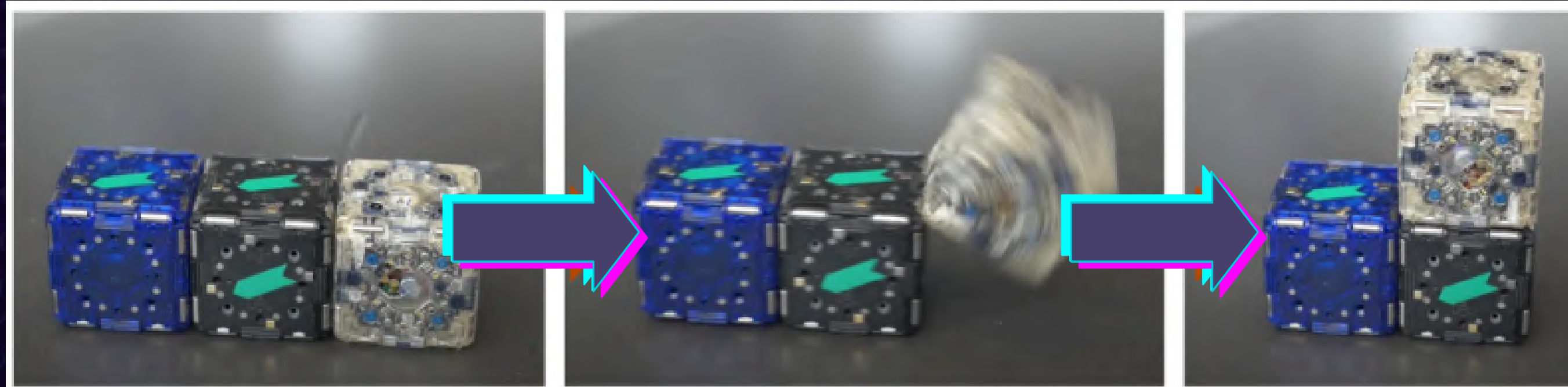
RoboSim



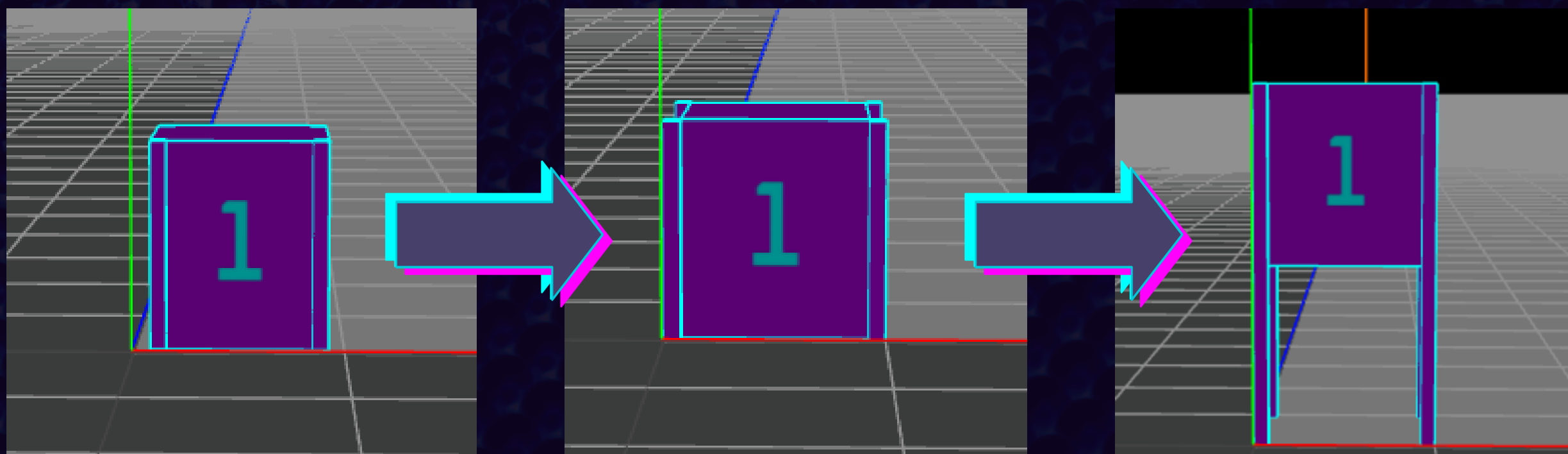
- Self-moving, self-stacking robotic boxes
- RoboSim is a simulation platform for algorithm developers
- Ranks all algorithms on leaderboard based on time/moves/energy used
- No human interaction needed!

BOX TYPES

TYPE 1



TYPE 2



USER STORIES



WAREHOUSE MANAGER

As a warehouse manager, I want to optimise the storage and retrieval of goods, so that I can increase efficiency and reduce the overheads of inventory management.



ALGORITHM DESIGNER

As an algorithm designer, I want to develop and benchmark routing algorithms with an intuitive API, so that I can streamline testing and optimisation of robot navigation in 3D space.



FUTURE CONTRIBUTOR

As a future contributor, I want support with understanding the existing codebase and project so I can get started quickly.

DEMO!

A horizontal, glowing purple oval shape is positioned directly below the word "DEMO!". It has a soft, ethereal appearance with a gradient from a lighter purple in the center to a darker purple at the edges.

TECH STACK

Web Client



TS TypeScript



Database



Server

TS TypeScript



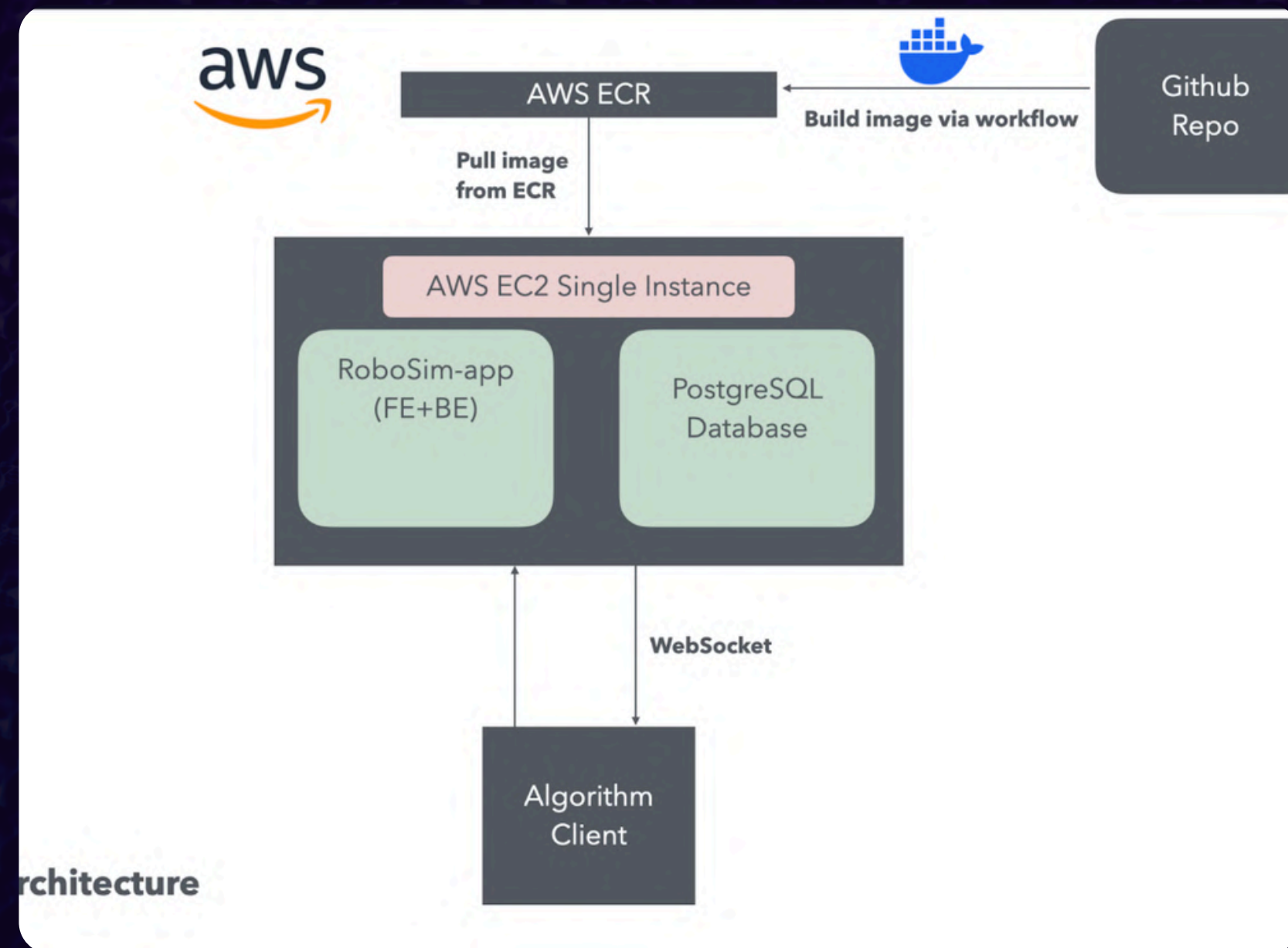
Algorithm Client



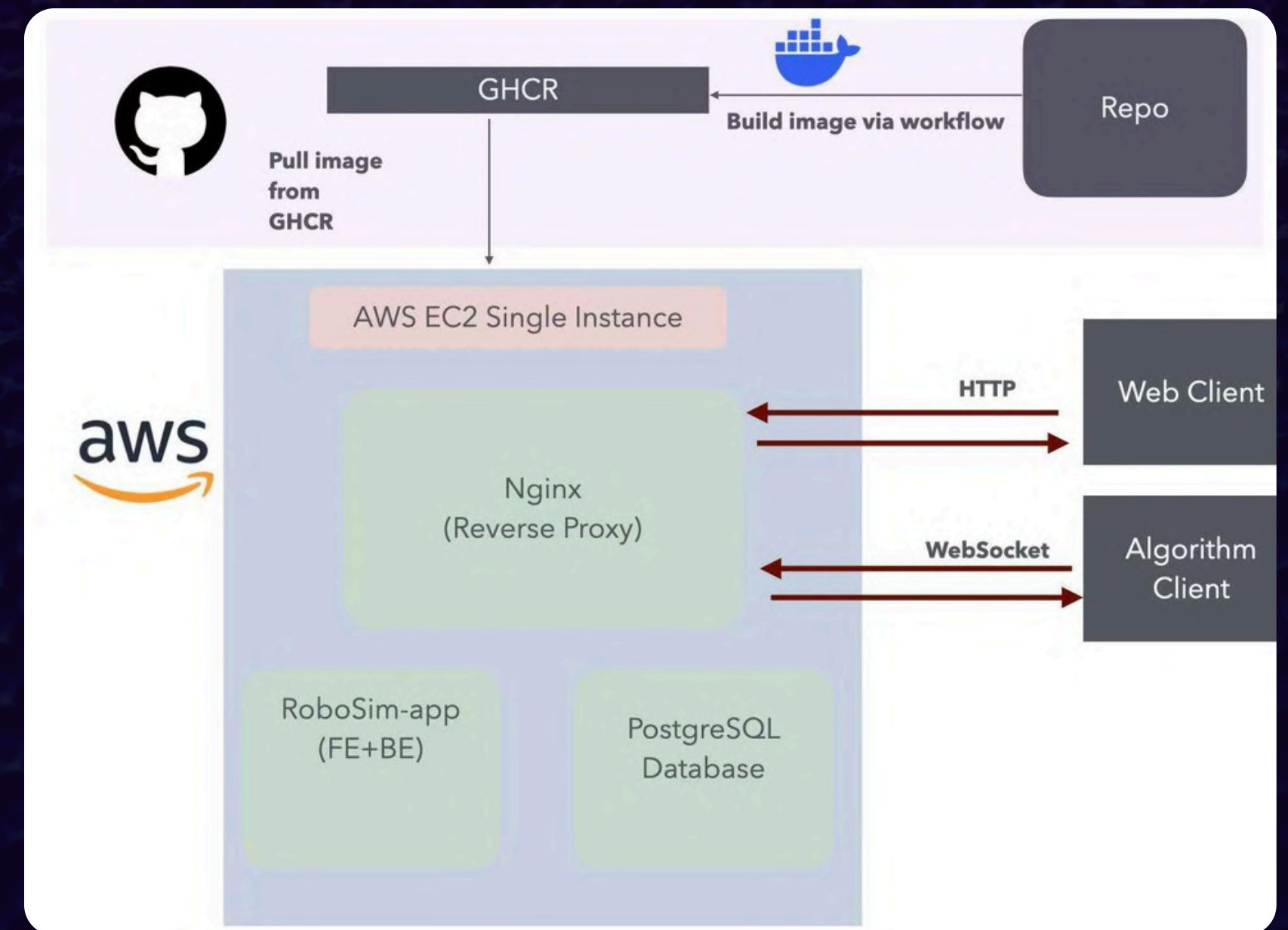
ARCHITECTURE DIAGRAM

Our website:
<http://13.49.70.251>

Before:



After:



CI/CD

Last Year

- **124** unit tests
- Testing *less* functions

124 passing (3s)

```
===== Coverage summary =====
Statements   : 89.51% ( 3713/4148 )
Branches     : 81.8% ( 454/555 )
Functions    : 96.81% ( 152/157 )
Lines        : 89.51% ( 3713/4148 )
```

This Year

- **188** unit tests
- **More** functions and lines of code tested

188 passing (3s)

```
===== Coverage summary =====
Statements   : 89.69% ( 4075/4543 )
Branches     : 84.2% ( 725/861 )
Functions    : 95.57% ( 281/294 )
Lines        : 89.69% ( 4075/4543 )
```


TESTING DAY

Users filled in a survey and we analysed the feedback and broke it down into a table so we knew what we needed to focus on

Feedback	Resolution	Justification	Area	Ease	Priority
"Could add directions (north/south or door/entrance)"	Add entrance/exit zones to the warehouse simulation for boxes to be moving towards.	Makes the playground a more clear and real representation of a warehouse	Algorithms	Hard	High
"Maybe add some shelves with boxes to mimic a warehouse"	Makes it a clearer representation of a warehouse.	Not that type of representation of a warehouse as boxes are designed to stack on top of each other.	Frontend	Medium	Low
"Depending on use cases, maybe a <u>first person</u> view or a more warehouse like background would indicate more context to the user"	Makes it a <u>more clear</u> representation of a warehouse.	Gives the user context that the playground area is where they simulate the warehouse. <u>However</u> system is currently <u>first person</u> view.	Frontend	Medium	Low
Add "an optional tutorial" when simulating the boxes	Useful to guide the system through how to use the platform.	Means users know how to navigate and do stuff when opening the platform for the first time.	Frontend	Hard	High

Include "clearer instructions on how to move boxes"	Allows the user to easily navigate the system	Will improve the usability of the system	Frontend	Medium	High
Adding functionality of "being able to place blocks not only in the origin square but also any other square, maybe and insert mode for adding boxes and a moving mode for moving boxes"	Would mean when adding <u>boxes</u> they're not just stacked on top of one another which leads a new user to confusion about how to unstack and move etc	Will really improve the ease of use.	Frontend	Medium	Medium
"Ability to replay simulation without reloading the page"	Means replay will function correctly	<u>Definitely</u> a bug to fix, improves usability	Backend	Medium	Medium
"Maybe different shaped boxes so not all cubes"	Additional feature that could be simulating a warehouse that isn't using the MIT box types that <u>Robosim</u> models	Could be inefficient if we're modelling shapes that don't stack optimally, defeating the point of optimally moving the boxes to target locations	Backend/Frontend	Hard	Low
Being able to drag, compress and expand the instructions blob	Improves usability	Currently blocks the playground area meaning you <u>cant</u> work through <u>it</u> side by side	Frontend	Medium	Medium

or algorithms when simulating the boxes"	use	have to navigate back to preview other algorithms or replays			
"Having a FAQs question symbol across all pages"	For bugs/how to use the page	Will improve usability	Frontend	Easy	Low
"Including breakpoints in runs of algorithms"	For developers to be able to debug when an algorithm gets stuck and be able to see the state it stops at	Makes it easier for algorithm designers to simulate and improve their algorithms	Backend/Frontend	Medium	Low
"State-saving on the website"	Adding a "back button" and not resetting/refreshing the page when we go back	When in the replay page - when we click back, it should persist with the last tab we were at	Backend/Frontend	Medium	Medium
<u>"clearer instructions"</u>	Will fix the lack of clarity when learning to use the system	Something we need to improve	Frontend	Easy	High
"Difficult to understand for a new user what's going on. Would be nice to see some information running because they were very short and abruptly stopped."	Adding some kind of feedback on the screen that the simulation has finished	Would be an easy addition to the system to improve usability	Frontend	Easy	Medium
"I believe something to work on is how visible the boxes are when they are hidden behind"	Not sure how to fix yet but good thing to work on	Fixes error of box lines being in front of target box in transparent mode	Frontend	Medium	Medium
"Maybe more description is needed for how and why to use this very	Add to main page context behind Robosim	Will highlight how impressive <u>Robosim</u> is and the motivation behind it	Frontend	Easy	Low

ITERATION 2: TESTATHON!

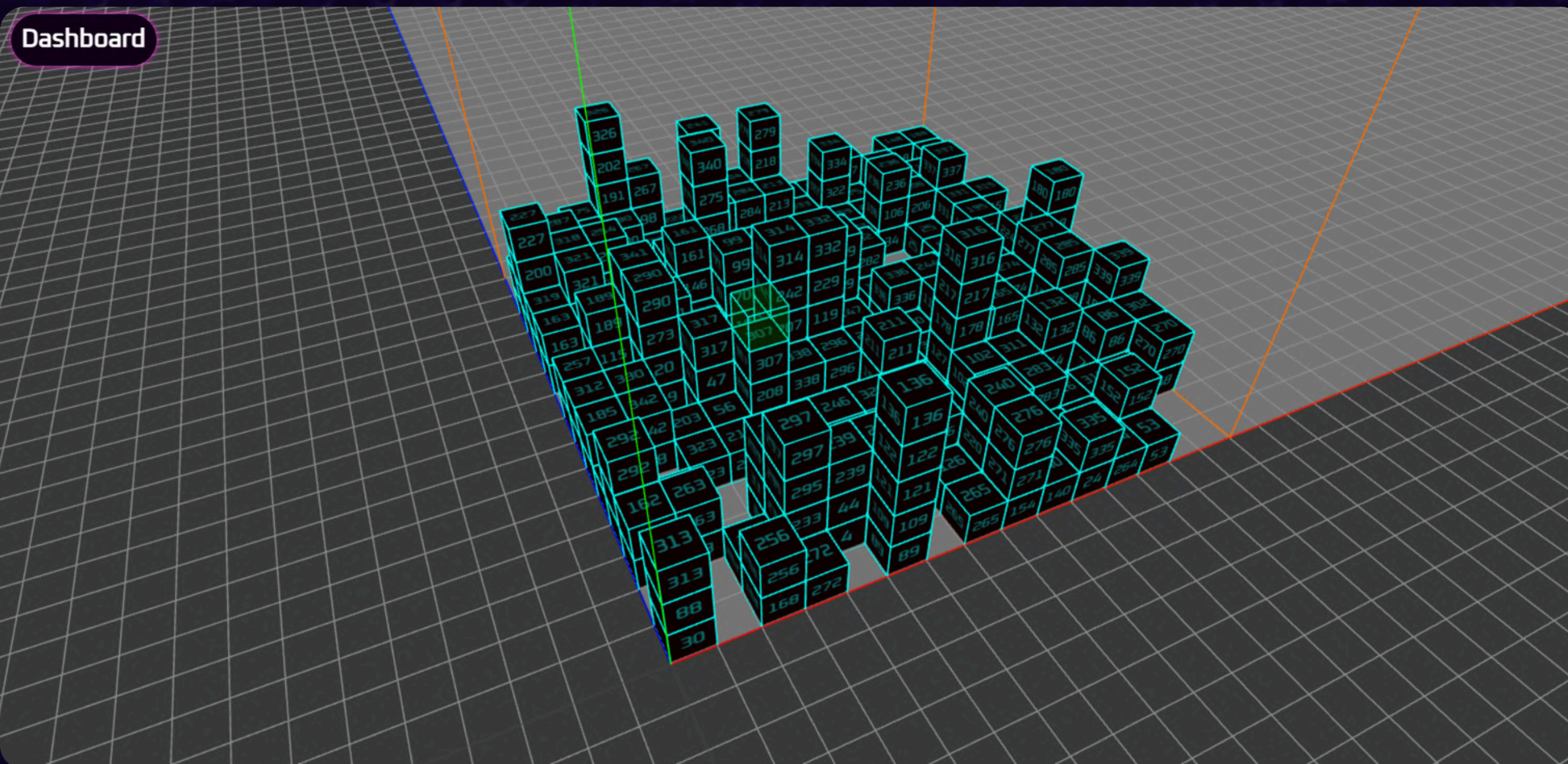
Think-aloud testing with users to evaluate how the users approach the system and see where they got stuck and confused

Testathon Feedback 04.03.26					
Feedback	Resolution	Justification	Area	Ease	Priority
"The playground is quite difficult to navigate, not too sure how to control it or complete actions, say move a box around."	Make more clear, concise instructions	Will make it easier for users to know what to initially do as most get stuck when opening the playground	Frontend	Easy	Medium
"The instructions on the playground weren't too clear"	Break the instructions down and make them more concise	Will make users less overwhelmed when first using the system	Frontend	Easy	Medium
"I think the buttons in the controls page look very similar, maybe make them different sizes depending on the importance of use. E.g Add box should be bigger and bolder"	Make more important buttons (add and remove a box) larger so users know where to look	Will make the system less complex as users currently open it to many boxes and not sure where to start	Frontend	Easy	Medium
"Depending on use cases, maybe a first person view or a more warehouse like background would indicate more context to the user"	Put the playground in first person and add more warehouse like features	Definitely a good addition that we are currently working on with the warehouse features (exit/entrance zones) being made – just requires a lot of additions to backend too so boxes move around/towards warehouse features	Frontend/backend	Hard	High
"Didn't understand the 'travelling box' option"	Rename 'travelling box' to something else	Good change to make as users were confused what	Frontend	Easy	Medium
intense it all being opened up initially"					
"When I press generate graph, it doesn't show the correct axis that I select"	Fix the error that is causing this	Will mean graph generation works correctly which is something we ideally want	Backend	Medium	High
"I can't undo the camera features once I've applied them"	Make it so features are removable when running scenarios	Something we should try and fix	Backend	Medium	Medium
"The 3 initial hovering boxes aren't clickable but look clickable"	Make boxes lead to the pages they say on them rather than just hover	Has caused a lot of confusion with users when initially on dashboard	Backend	Easy	Medium

	that is understandable by someone non-technical	it was referring to, but also our general stakeholders might understand the references more			
"Make play side on the left larger potentially as people are looking at 'traveller box' first"	Make play button larger	Will mean the more important features are more visible	Frontend	Easy	Medium
"Past algorithms on the dashboard don't stay open when you come back to the page"	Make sure algorithms that have been opened remain open	Will make it easier for algorithm developers to go back to their work when navigating the system	Backend	Medium	Low
"Now we have the instructions open first, I didn't notice the list of 'my algorithms'"	Maybe we just close the instructions but make the button to open it bigger or open the first step only	Will make the playground less overwhelming when initially opened	Frontend	Medium	Medium
"When I pressed 'Add Box' I was unable to remove them after for a while"	Make the 'remove box' button bigger/more visible somehow	Will get rid of the confusion when a user tries to navigate the playground	Frontend	Easy	Medium
"When removing box weird transparent thing left behind"	Get rid of it when boxes are removed	Will look better and cause less confusion	Frontend	Medium	High
"I don't understand how to place boxes alongside each other"	Improve our instructions so they're more clear	Will make it easier for a user initially using the system	Frontend	Easy	Medium
"I like that the instructions are clear"	Make the instructions clearer	Will make it easier for a user initially using the system	Frontend	Easy	Medium

CHANGES
WE'VE MADE!

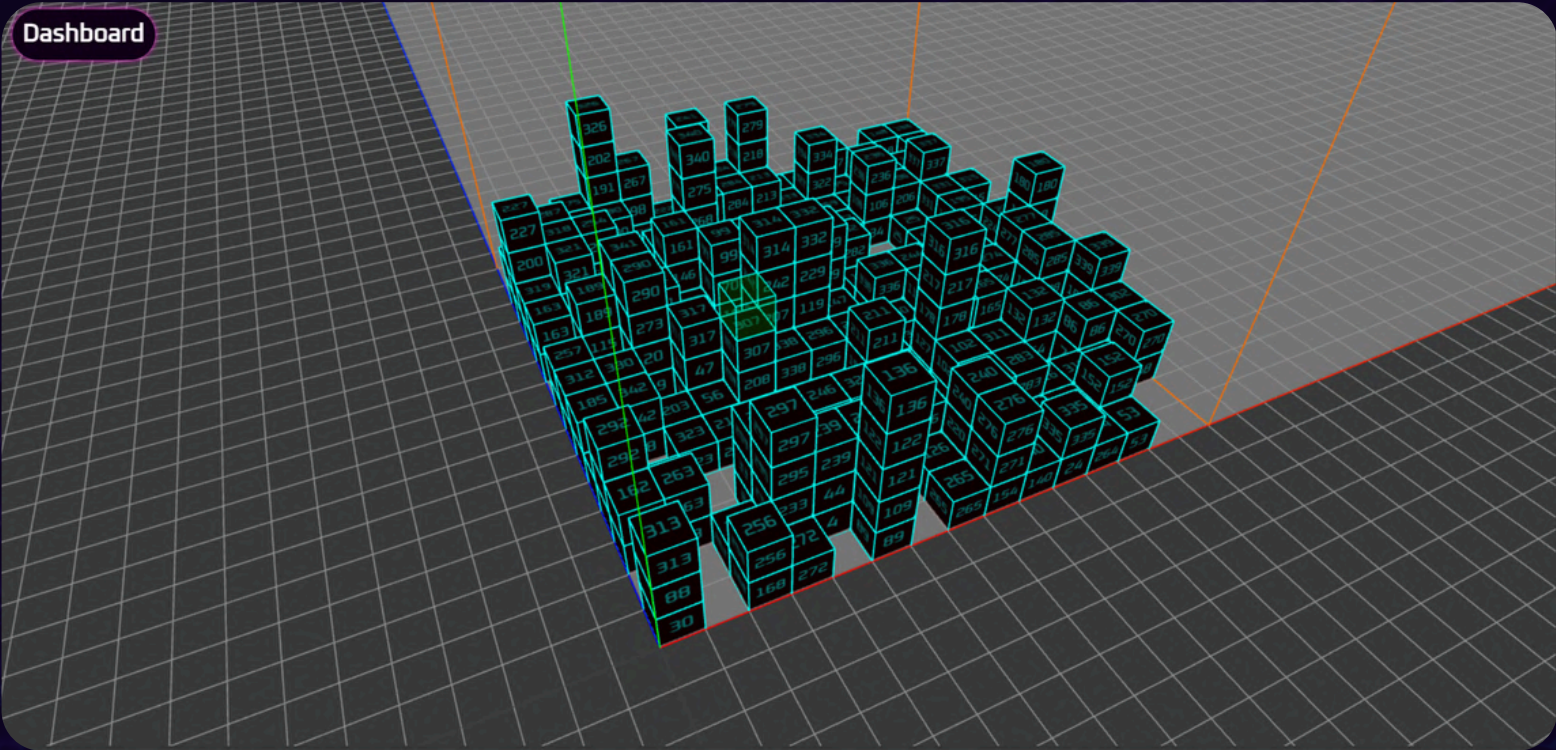
PROBLEM: REPLAY/LIVE VIEW



- System immediately started running when the user entered the replay
- Users needed to manually control the camera to track the box
- Couldn't always tell which box we were focusing on
- Couldn't pause or restart the run

SOLUTION 1:

Before



Select traveller box to target

Traveller: 2619

Traveller: 746

Traveller: 1765

Traveller: 2619

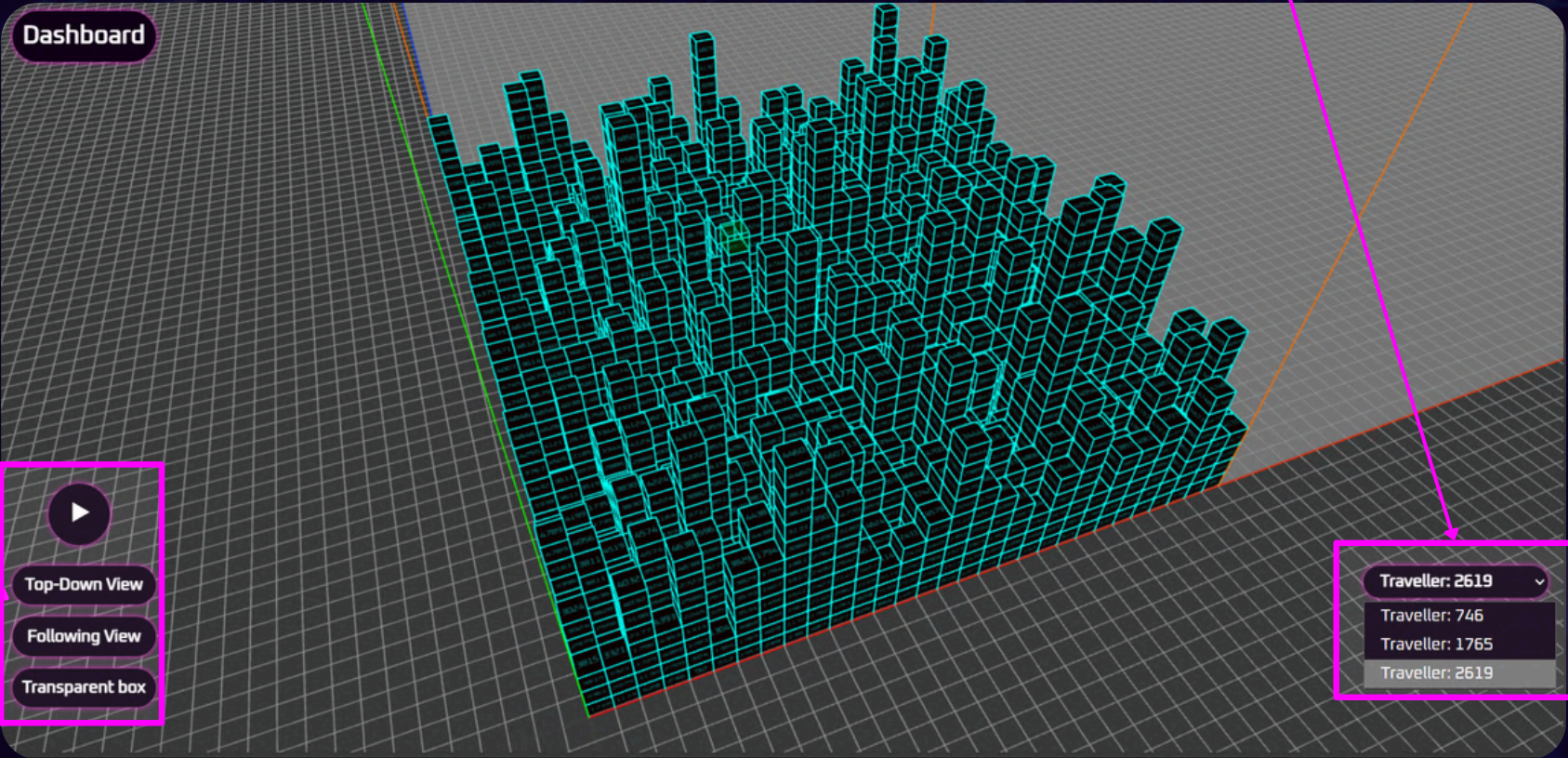
After

Top-Down View

Following View

Transparent box

View mode



SOLUTION 2:

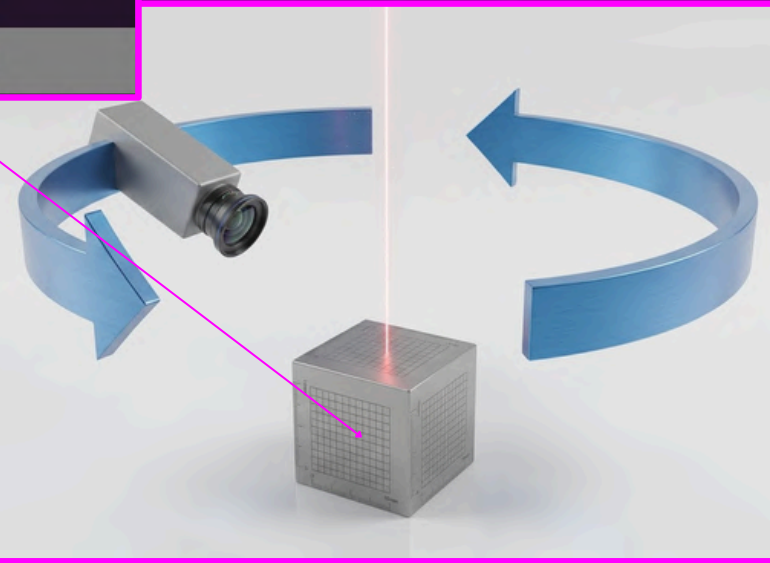
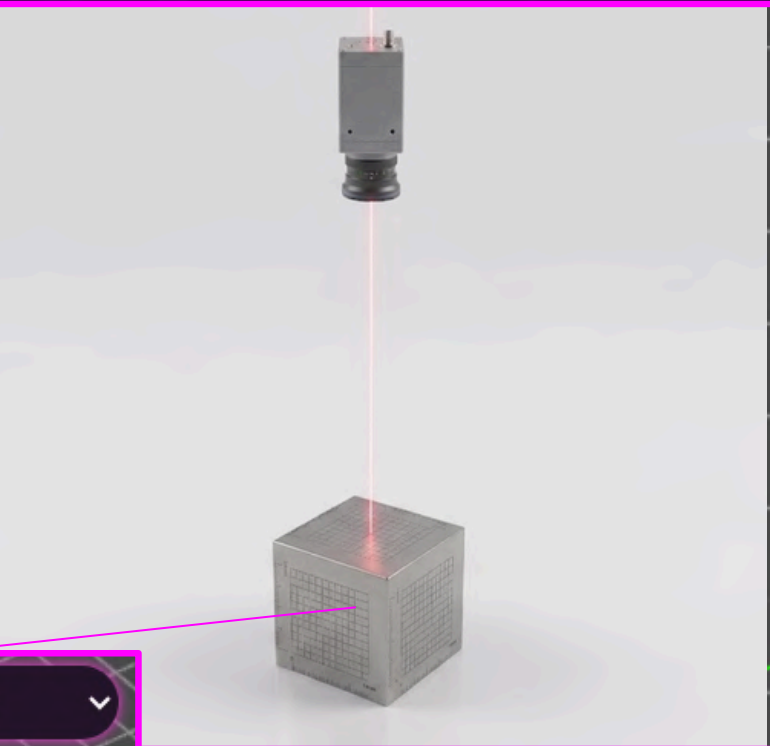
Traveller: 2619

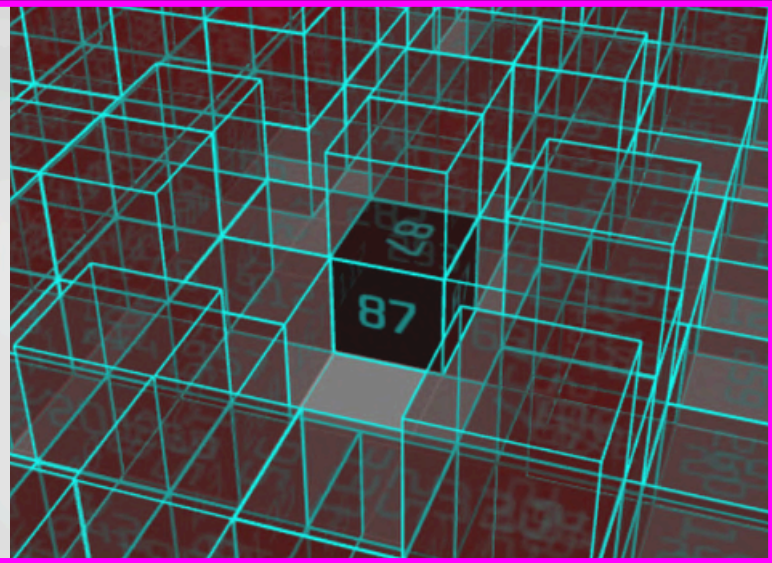
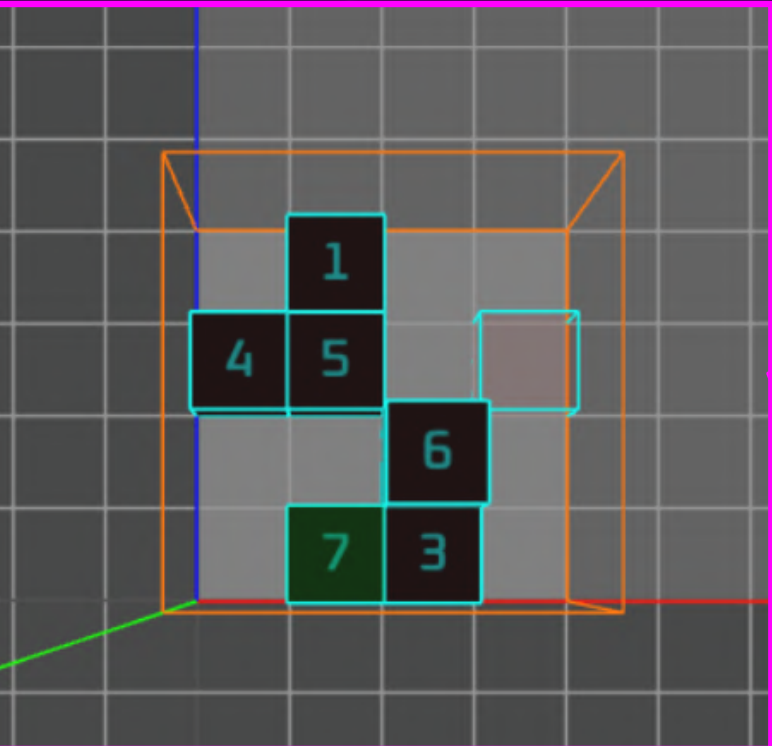
Traveller: 746

Traveller: 1765

Traveller: 2619

← Select Target ID





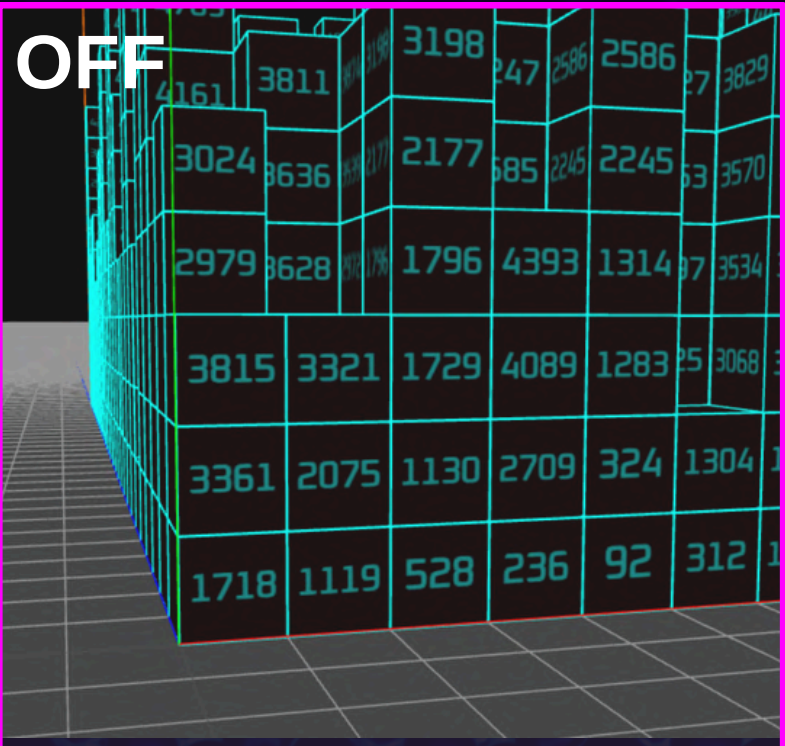
Play
Pause
Restart

Top-Down View

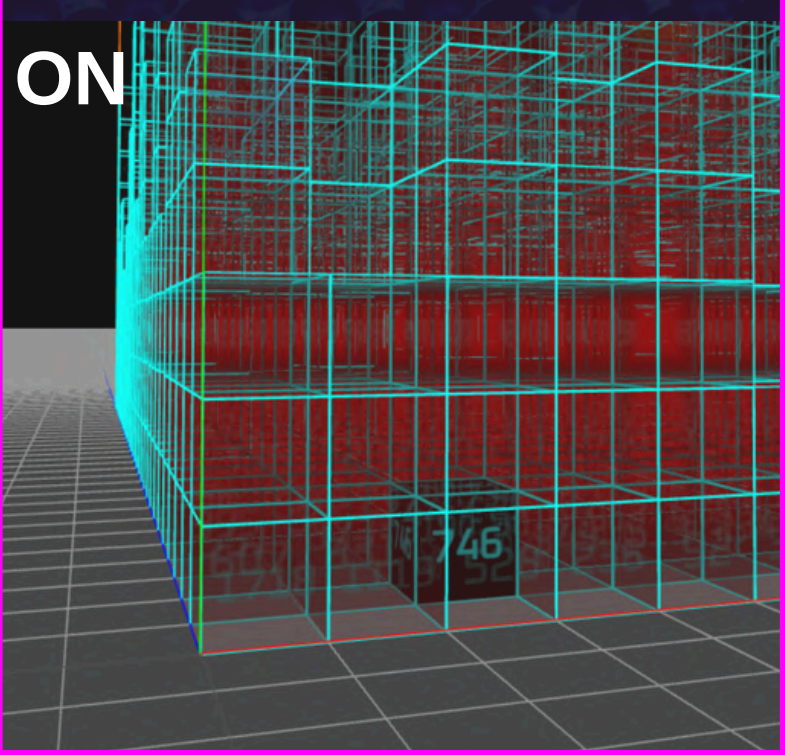
Following View

Transparent box

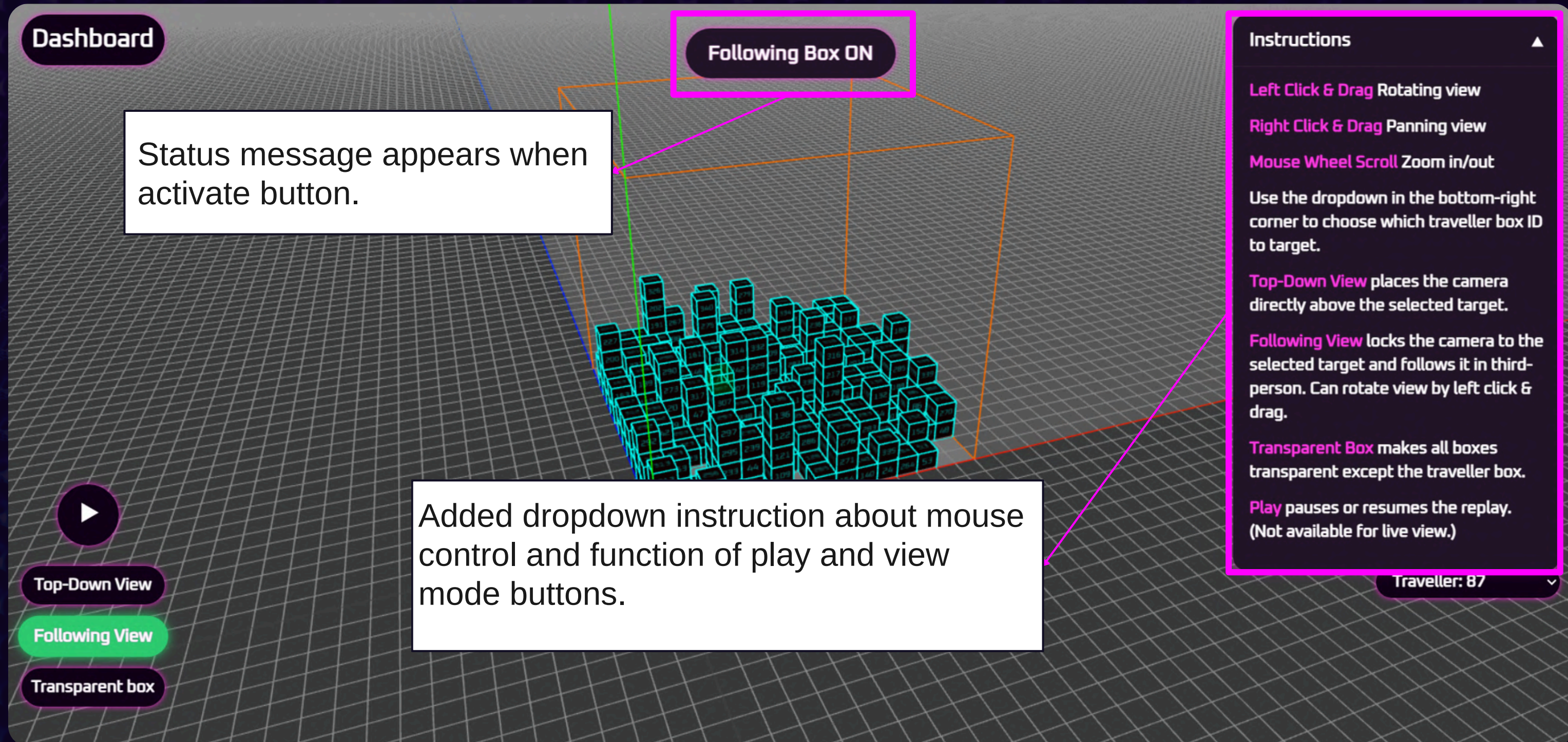
OFF



ON

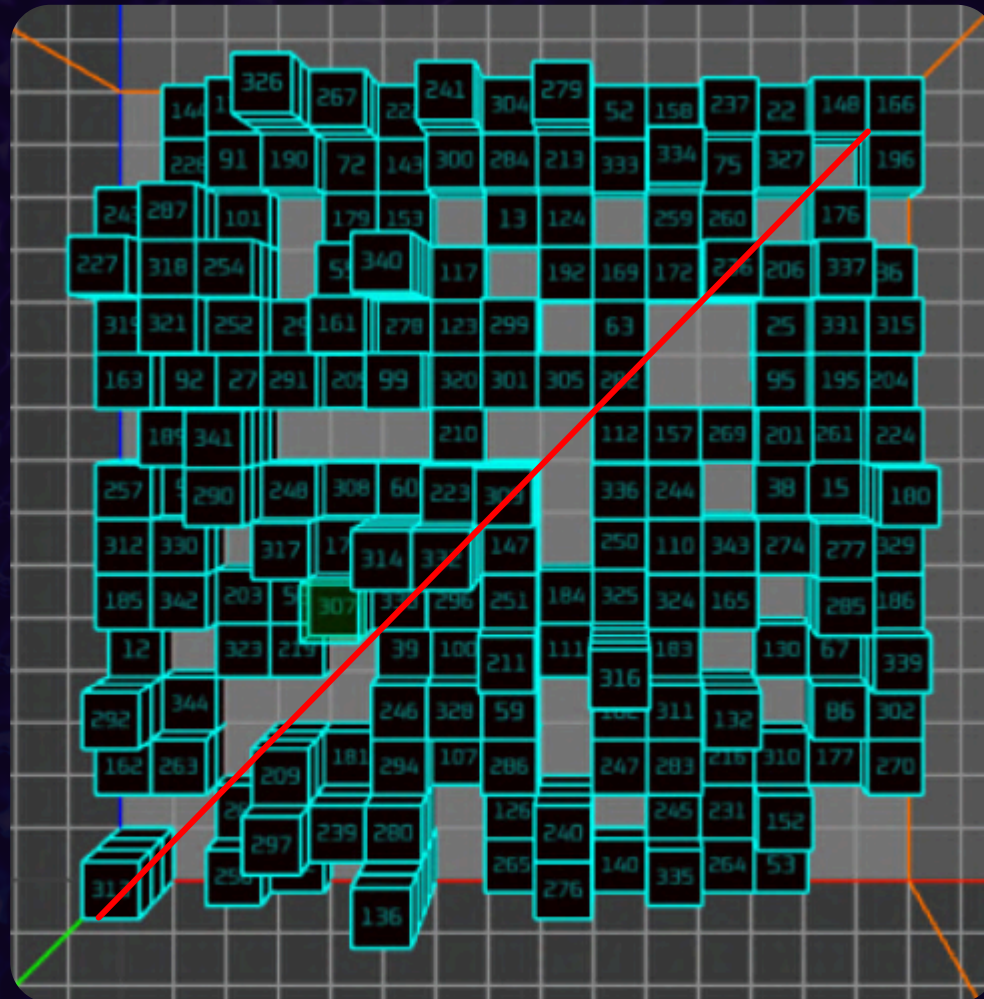


SOLUTION 3: INSTRUCTIONS ON REPLAY/LIVE VIEW



CLIENT FEEDBACK

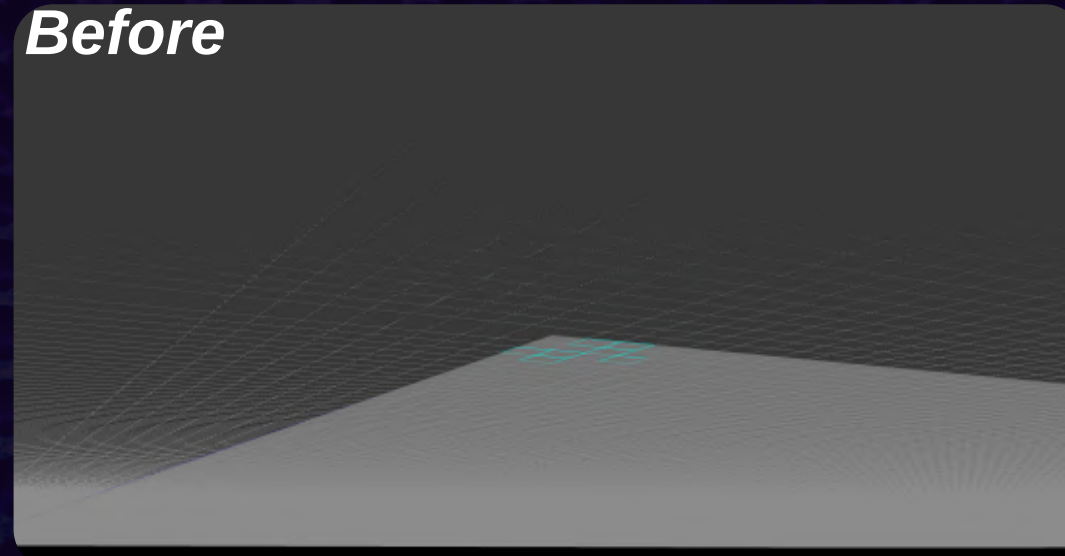
Zoom out distance limit



Calculate ratio of diagonal length then decide the zoom out distance limit

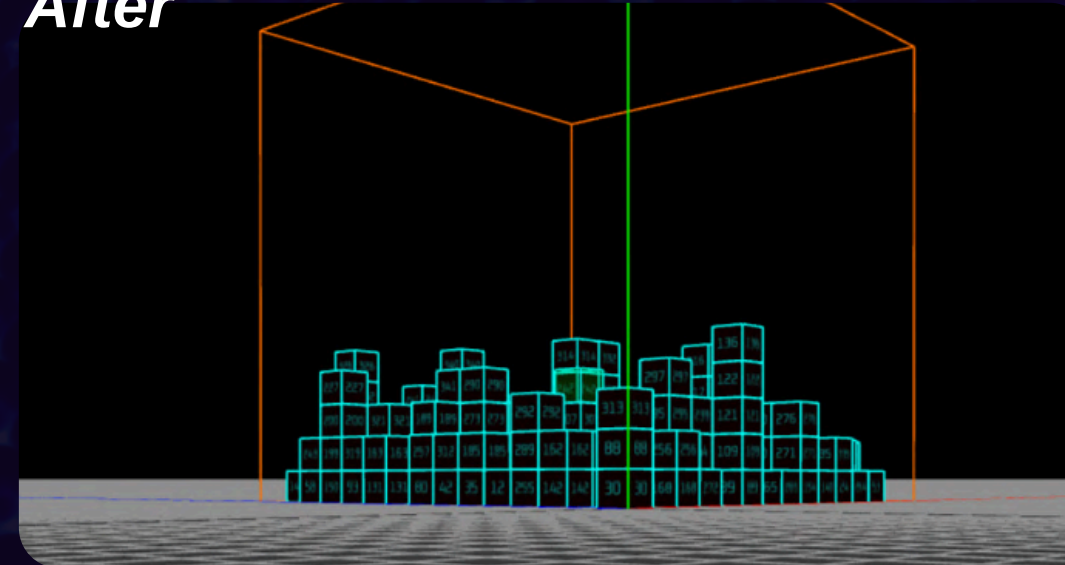
Floor collision

Before



Camera could go under the warehouse floor.

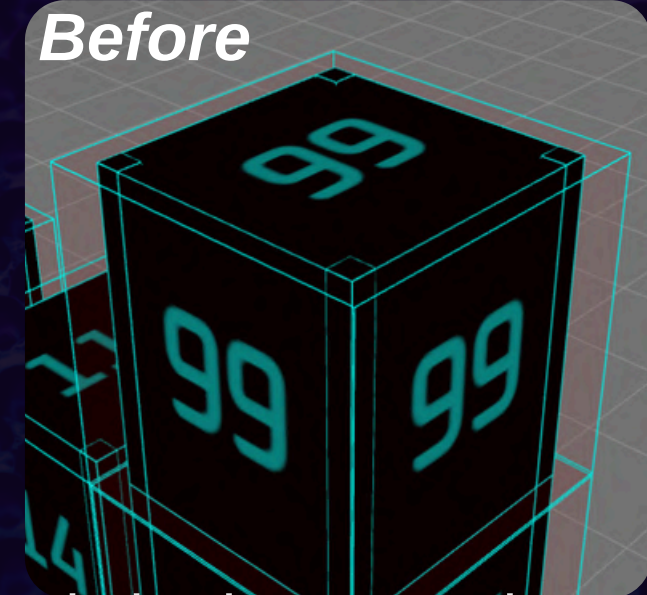
After



Camera cannot go under floor

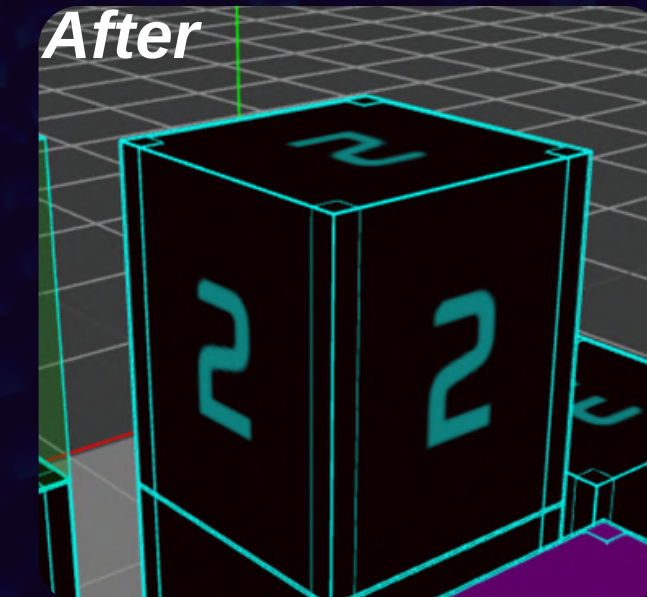
Reduce visual complexity of Type 2 box

Before



Red shade covered type 2 box

After



Red shade matches to type 2 box

DATABASE

Why change it?

- The current database was wiping the data every time it was deployed
- Inefficient to save algorithm data to analyse
- Researched what would be suitable solutions to fix this error



DATABASE

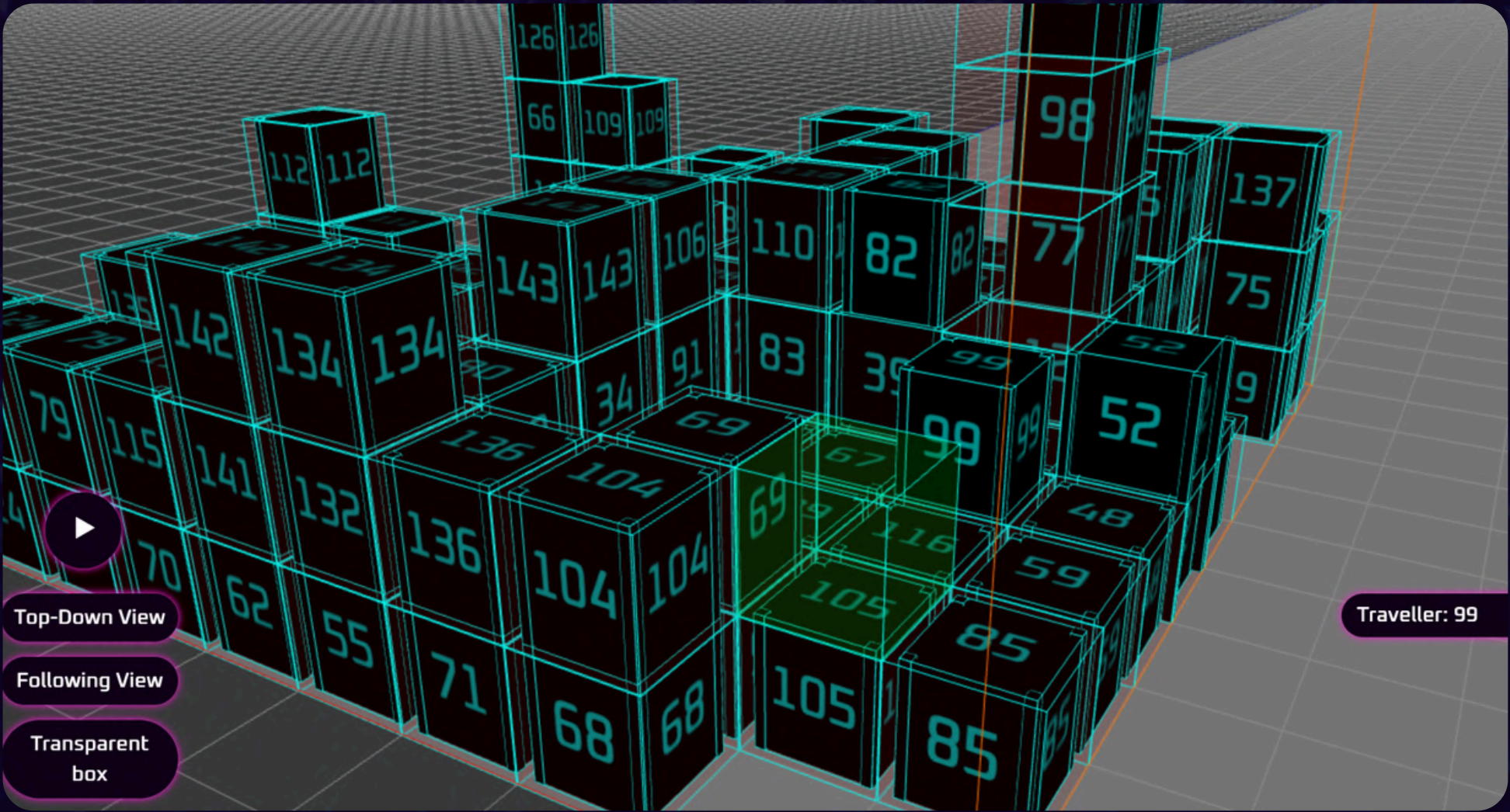
Challenges:

- Huge amount of work - took far longer than planned
- New to the codebase
- Bugs!!!
- Caused a huge amount of test cases to be broken - had to work into term 2 to fix



TYPE 2 ALGORITHM

- Able to solve all single-requirement scenarios
- Multi requirement scenarios passed off to handover



Before:

Algorithm Client Summary			
Name: Type 2		Author: Raymond Kellner	State: Running
Scenario	State	Energy Used	View
Scenario 11 - Type 2	Completed in 5.0s	5	View replay
Scenario 13 - Type 2	Pending	0	Follow run
Scenario 12 - Type 2	Completed in 4.0s	3	View replay
Generate Graph			

After:

Algorithm Client Summary			
Name: Type 2		Author: Raymond Kellner	State: Completed
Scenario	State	Energy Used	View
Scenario 11 - Type 2	Completed in 6.0s	5	View replay
Scenario 13 - Type 2	Completed in 14.1s	13	View replay
Scenario 12 - Type 2	Completed in 4.0s	3	View replay
Scenario 14 - Type 2	Completed in 17.1s	5	View replay
Scenario 15 - Type 2	Completed in 44.3s	61	View replay
Generate Graph			

DASHBOARD

- Transformed the dashboard into a landing page which introduces the project to new user
- Hid the algorithm runs behind a drop down menu to visually de-clutter

Playground Instructions Leaderboard

RoboSim

Your Algorithms

All Algorithms

No algorithms are currently running. See the Instructions page to run your own.

Hide Algorithm Runs ▲

Your Algorithms

All Algorithms

✖ Type 2 - Raymond Kellner 2 completed
6 skipped
5 failed ⋮

✖ Fill-Remove - Raymond Kellner 9 completed
0 skipped
1 failed ⋮

RoboSim

Playground Instructions Leaderboard ⚙

RoboSim

Algorithm Testing & Warehouse Simulation

Design, test, and compare algorithms in a 3D simulated environment.
Write your algorithm in Python, connect it to the server, and watch the boxes navigate scenarios in real-time.

Get Started

Develop & Test Algorithms

Our built-in Algorithm Client package automatically handles all of the communication between the backend and server.

Live Replay/Simulation

See how your algorithm performs in real-time, or watch replays of past runs, with multiple perspective options available.

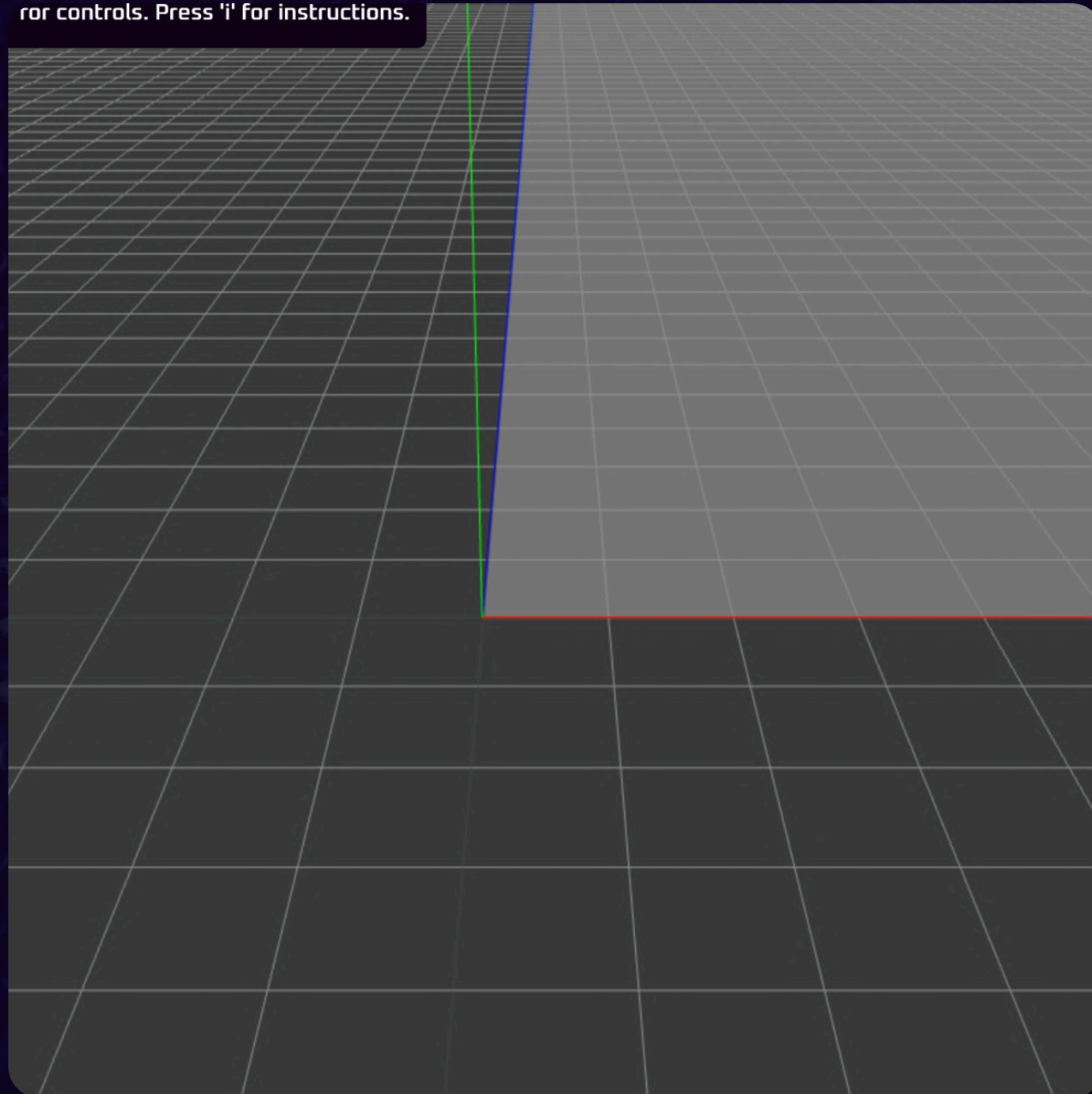
Compete & Compare

Use the leaderboard and graph features to compare the performance of multiple algorithms side-by-side, filtering by scenario, algorithm type, and metric.

PLAYGROUND

- Visually de-cluttered by removing redundant controls
- Auto-load all visual elements for more intuitive design

ror controls. Press 'i' for instructions.



PLAYGROUND

- Moved instructions box into scrollable menu
- Added pop-up upon loading playground
- Scenario wizard which handles scenario generation

Welcome to the **Playground** - in here, you can add boxes to create warehouse scenes and scenarios.

Press **C** to toggle controls

Press **Tab** for help

Dismiss

Name Your Scenario

Scenario name:

1 target:

Box 2: (18, 0, 3) → (1, 1, 1)

Back

Download

Insert Mode

Hold **Space** to enter Insert Mode. While held:

- **Left click** to place a box at the cursor
- **Right click** to remove the targeted box

Camera controls are locked during Insert Mode.

Forklift Mode

In Forklift Mode, you can move a stack of boxes by selecting the bottom box (box on the ground), simulating a forklift. This makes it easier to build configurations of boxes.

Note: When building algorithms to move boxes, you cannot move stacks of boxes. This is purely to help with building scenes in the Playground.

Toggle with **Shift** + **F**

Rules & Restrictions

- You cannot move a box if it has boxes on top of it (except if moving the bottom box of a stack in **Forklift Mode**)
- You cannot move a box if the **height difference** between the current box position and new box position is greater than 1.
- You cannot remove a box if it has boxes on top of it.

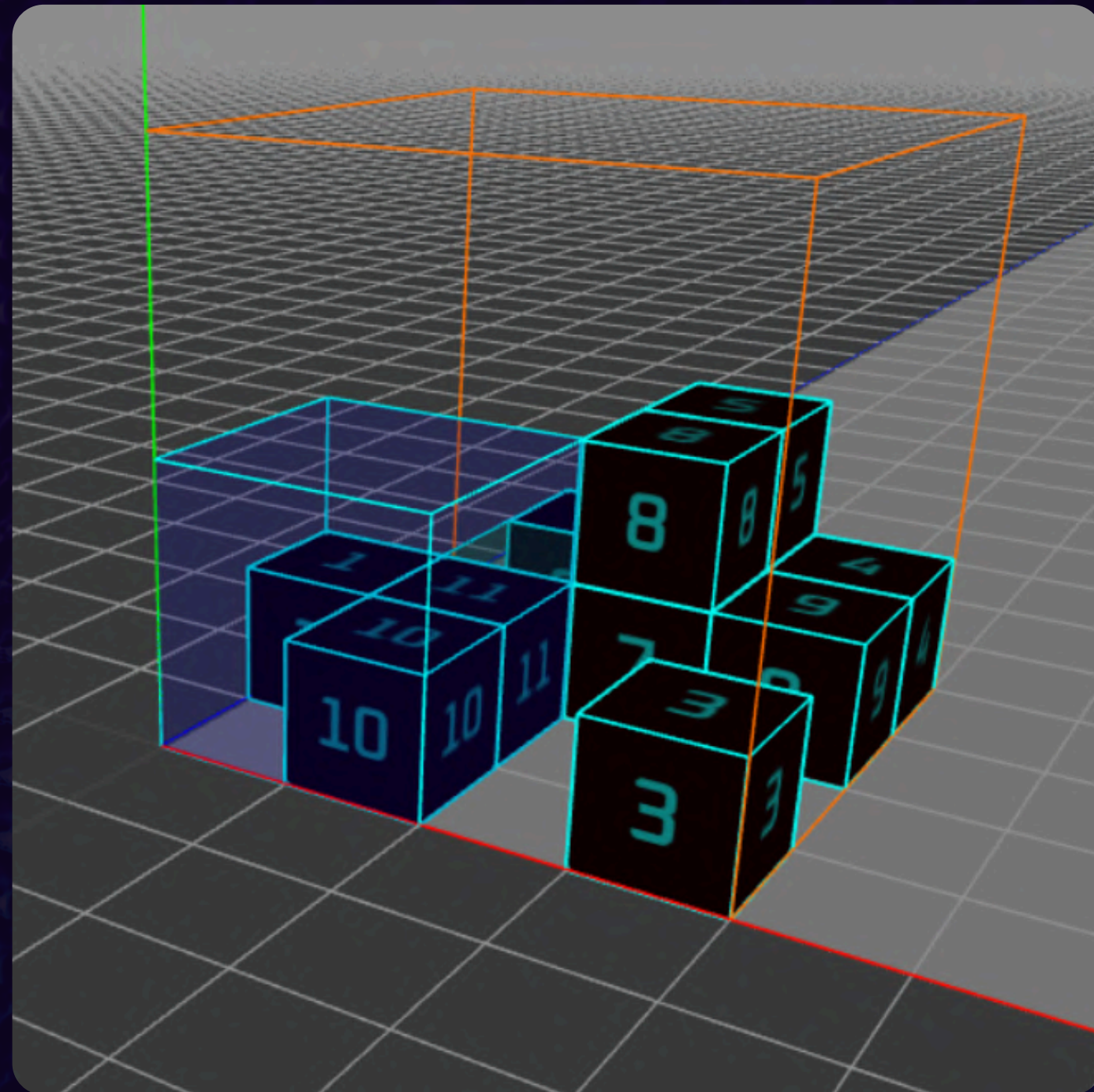
Scene Controls

Scenarios

A **scene** is a layout of boxes. A **scenario** adds target locations for selected boxes, creating a challenge for your algorithm to solve. Use the scenario wizard to create scenarios from the current scene.

Scenario Wizard

EXIT/ ENTRANCE ZONES



- Aiming to make a more realistic warehouse
- Completed frontend & some backend!
- Future use: algorithms using them, boxes coming in and out

ENERGY USAGE TRACKER

Playground Instructions Leaderboard

Algorithm Client Summary

Name: A* Client

Author: Admin

State: Running

Scenario	State	Energy Used	View
Scenario 1 - Type 1	Completed in 2.0s	6	View replay
Scenario 2 - Type 1	Completed in 4.0s	8	View replay
Scenario 3 - Type 1	Completed in 5.1s	12	View replay
Scenario 4 - Type 1	Completed in 15.5s	31	View replay
Scenario 5 - Type 1	Completed in 10.1s	22	View replay
Scenario 6 - Type 1	Completed in 1.1s	1	View replay

- Based on feedback from our MVP
- Allows users to evaluate efficiency of algorithms, not just the speed
- Very insightful new metric for our benchmarking

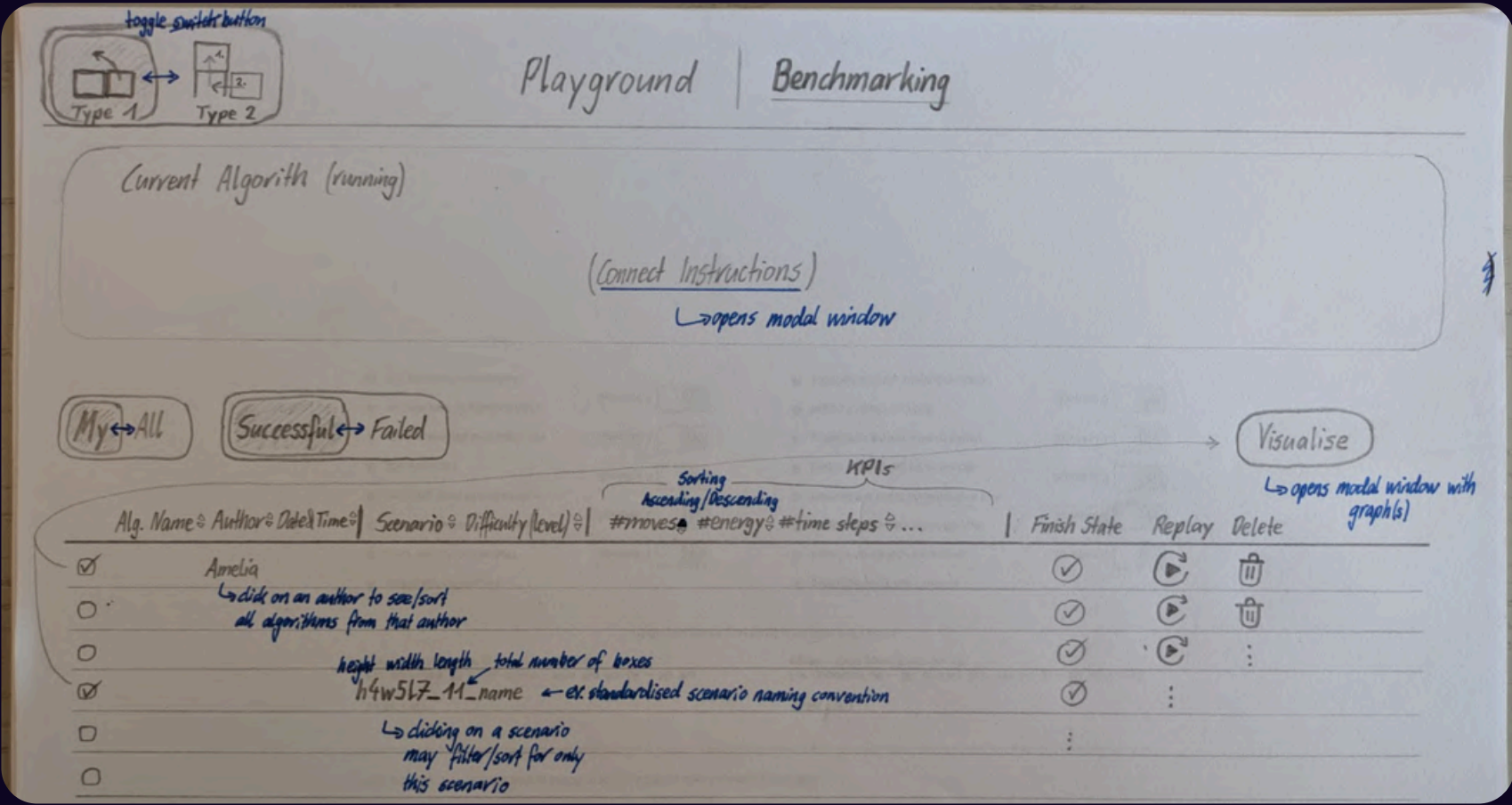
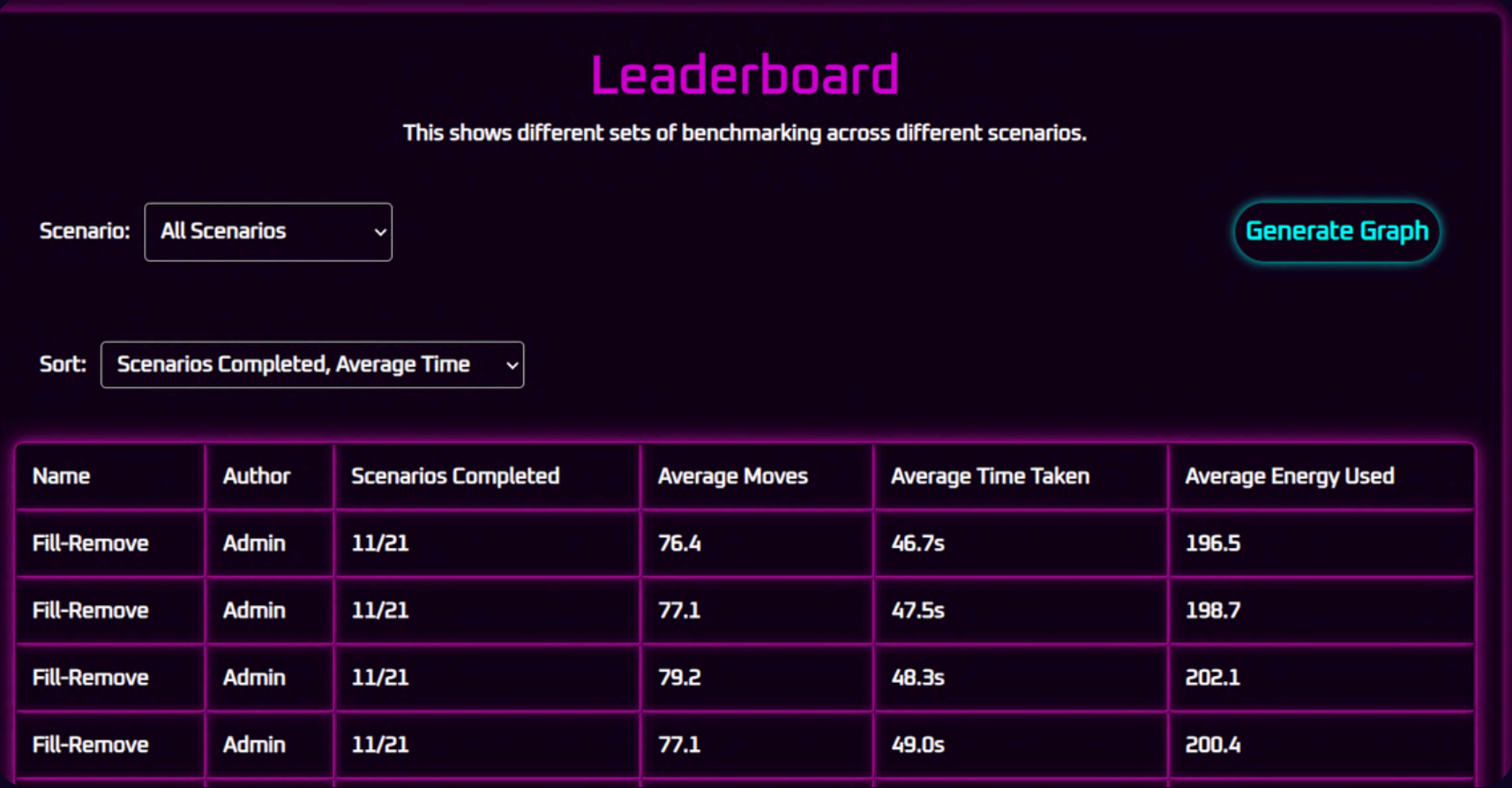
BENCHMARKING PAGE

The old leaderboard page:

Didn't include many metrics or let the user sort or filter algorithm runs.

Marius' sketch showing what he wanted us to create:

He wanted a page that gave the user much more insight into their algorithm runs.



NEW BENCHMARKING PAGE

Switch to Type 2

Generate Graph

Current Algorithm (running)

Name: A* Client

Moves: 0

Status: ▶ Running...

Scenario: Scenario 5 - Type 1

Energy: 0

How to Connect ▾

Live panel displaying algorithm runs in real-time

Current Algorithm (running)

✓ Run Complete! Updating table...

How to Connect ▾

✓	ALG. NAME ↓	AUTHOR ↓	DATE & TIME ↓	SCENARIO	# MOVES ↓	# ENERGY ↓	# TIME ↓	FINISH	REPLAY	DELETE
■	A* Client 917e48b3c543459eacc29eb3eb15540f	Freya	28/04/2026 14:56	Scenario 6 - Type 1	1	1	1298	✓	▶	🗑️
■	A* Client 917e48b3c543459eacc29eb3eb15540f	Freya	28/04/2026 14:55	Scenario 5 - Type 1	10	22	11190	✓	▶	🗑️
■	A* Client 917e48b3c543459eacc29eb3eb15540f	Freya	28/04/2026 14:55	Scenario 4 - Type 1	15	31	15953	✓	▶	🗑️

Leaderboard

This page shows different sets of benchmarking across different scenarios.

Switch to Type 2

Generate Graph

Current Algorithm (running)

Awaiting connection...

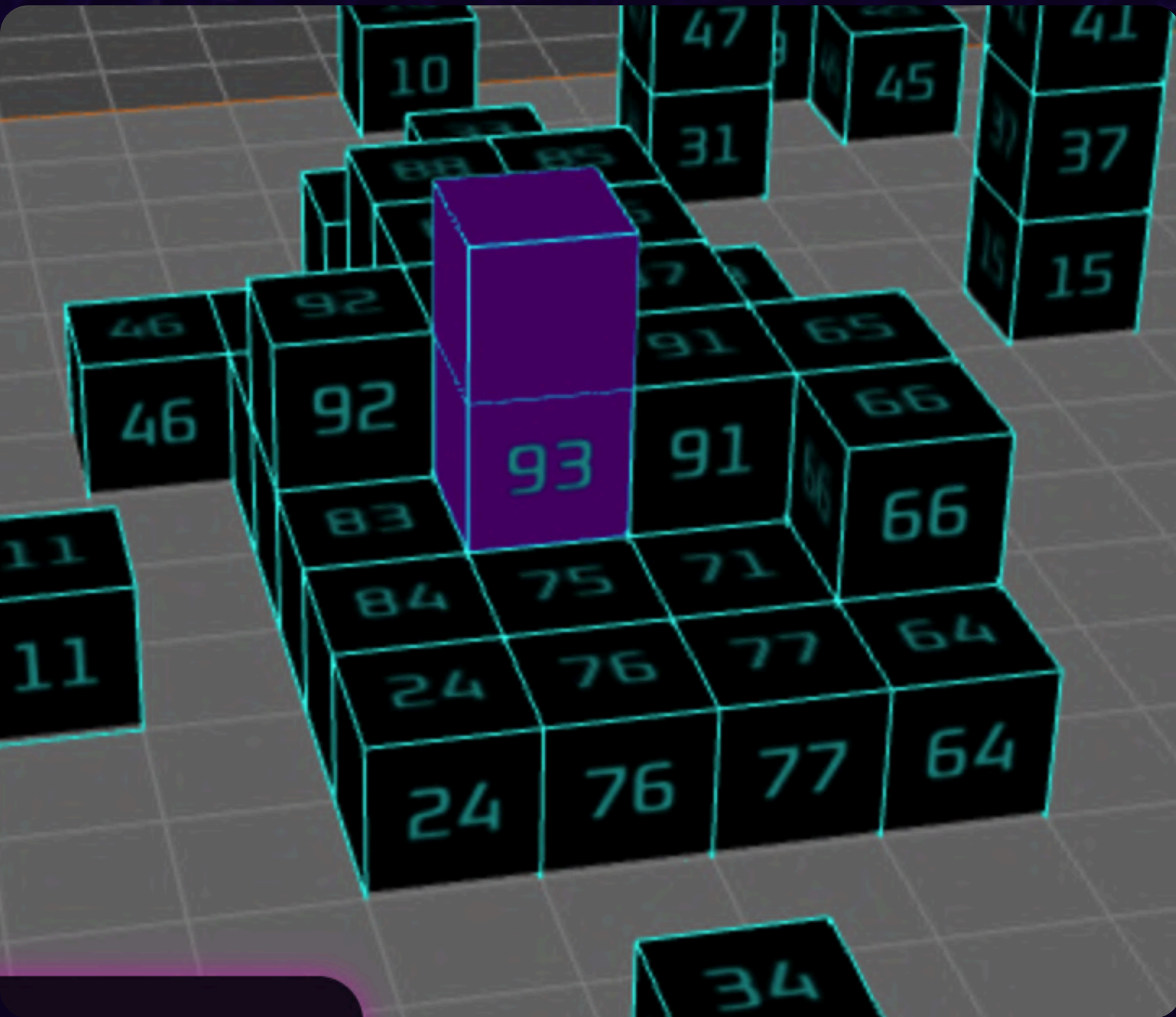
How to Connect ▾

My All Successful Failed

✓	ALG. NAME ↓	AUTHOR ↓	DATE & TIME ↓	SCENARIO	# MOVES ↓	# ENERGY ↓	# TIME ↓	FINISH	REPLAY	DELETE
■	A* Client 917e48b3c543459eacc29eb3eb15540f	Freya	28/04/2026 14:56	Scenario 6 - Type 1	1	1	1298	✓	▶	🗑️
■	A* Client 917e48b3c543459eacc29eb3eb15540f	Freya	28/04/2026 14:55	Scenario 5 - Type 1	10	22	11190	✓	▶	🗑️
■	A* Client 917e48b3c543459eacc29eb3eb15540f	Freya	28/04/2026 14:55	Scenario 4 - Type 1	15	31	15953	✓	▶	🗑️

Algorithm runs are now sortable by metrics and can be filtered by author, scenario, type of box and successful/failed runs

INSERT MODE AND DIMENSIONS



- Based on client feedback: much more efficient way of creating scenes and placing boxes
- Fully validates Type 1 & 2 box behaviour
- Able to change dimension size in real time, used to be completely static



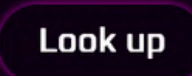
Insert Mode: Off

Insert Mode: On

SCENARIO GENERATION

- Previous year's scenario generation was unintuitive and had bugs
- Didn't produce .JSON files in the correct format
- Generated all boxes as requirements rather than just target boxes

Target 2 of 2

Box ID: 2  

Current location: (18, 0, 3)

Target location:

X: 12  Y: 4  Z: 6 

Back

Next

```
Name.json
{"id":38,"x":19,"y":0,"z":7},{"id":39,"x":18,"y":0,"z":6},{"id":40,"x":14,"y":0,"z":5},
{"id":41,"x":13,"y":2,"z":14},{"id":42,"x":8,"y":0,"z":3},{"id":43,"x":8,"y":0,"z":0},
{"id":44,"x":8,"y":1,"z":0},{"id":45,"x":13,"y":0,"z":18},{"id":46,"x":6,"y":0,"z":13},
{"id":47,"x":11,"y":1,"z":17},{"id":48,"x":7,"y":0,"z":1},{"id":49,"x":17,"y":0,"z":2},
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}
```


INSTRUCTIONS PAGE

Before:

- Only has a bit of explanatory text
- Documentation showed when user clicked the link

Algorithm Client Instructions

The Algorithm Client Package

The Algorithm Client Python package is a set of tools that allow you to test algorithms and send them to the server. See the [documentation](#) for more information, including how to install it.

Algorithm Client Instructions

The Algorithm Client Python package provides tools to test algorithms and submit them to the server.

You can read the required documentation directly below.

[Getting Started](#)[Submission Format](#)[FAQ](#)[Download current .md](#)

Documentation file explain how to run algorithms

Before we start the quick introduction of running algorithms,
We strongly recommend you to read the [detailed document](#).
By this file, you will figure out what is *algorithm client package*.

After:

- Documentation shows directly on instruction page
- More documentation shows
- Can download documentation from page
- We changed markdown files into HTML format by markdown parser

We used markdown parser

MORE INSTRUCTION CHANGES...

From our client feedback, we created a simple algorithm for users in the instruction page, which makes it more user friendly!

Quick start algorithm: Simple Algorithm

To help you make a quick start of our project, we provide a Simple Algorithm as an example:

[simple_algorithm.py](#)

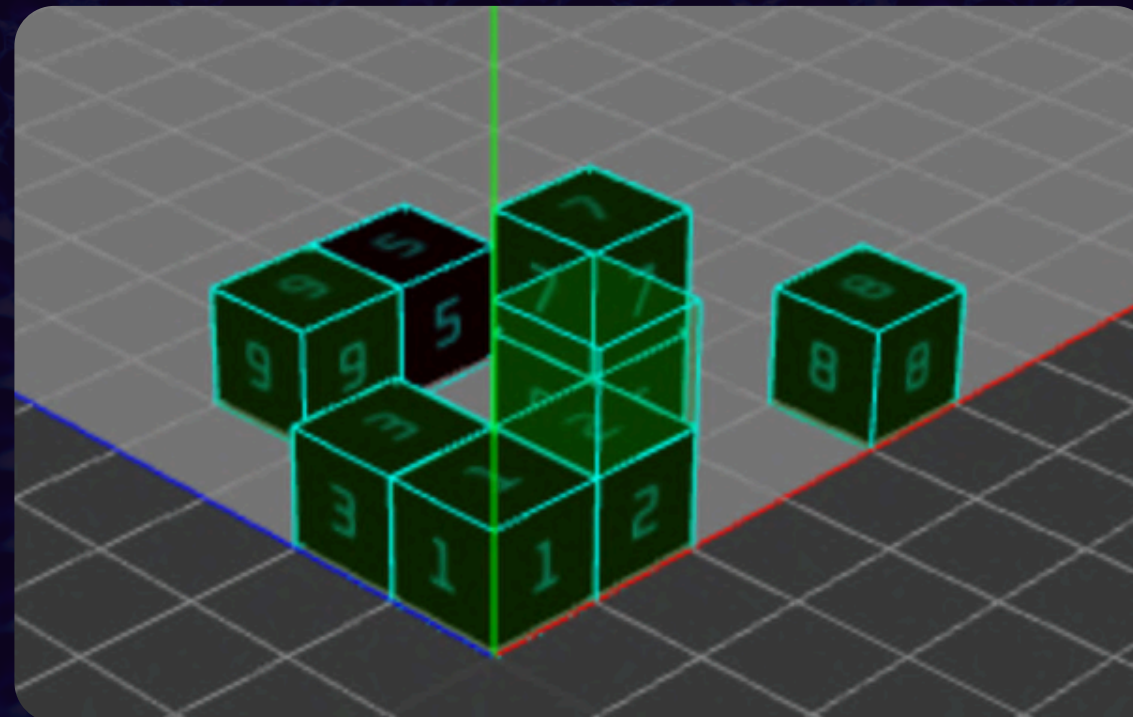
You could copy this python file and put it under *site-packages/*

Then you could cd into *site-packages/* and run: (If you name it *simple_algorithm.py*)

```
python simple_algorithm.py
```

After that, you will find your algorithm running in our website!

(the Simple algorithm introduction)



(the Simple algorithm scenario)



simple algorithm test mj - Marius

1 completed
0 skipped
0 failed



Could we have an easy algorithm to help users start our project?



TEAMWORK AND COLLABORATION

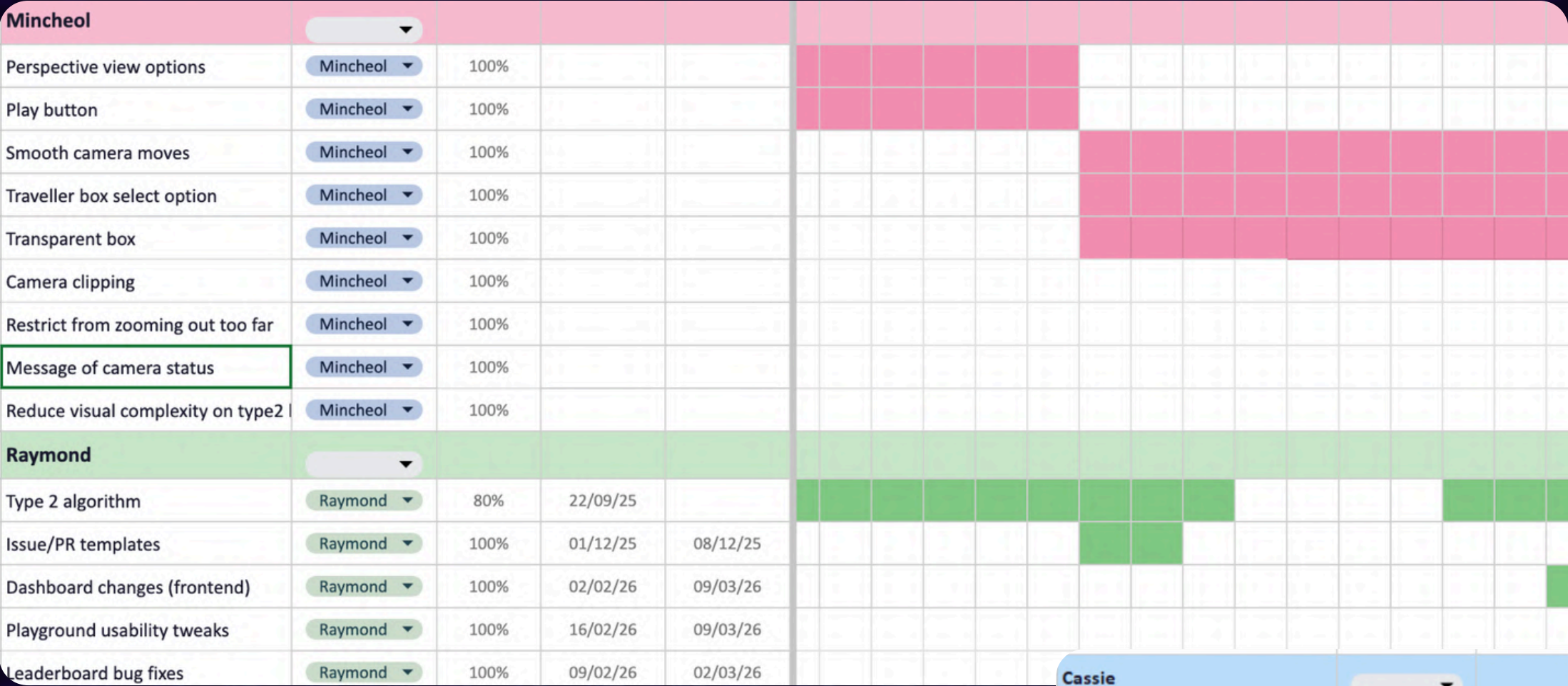
PROBLEMS

- Slow start in our first term
- Learning the huge existing codebase & setup issues
- Lack of communication in the team



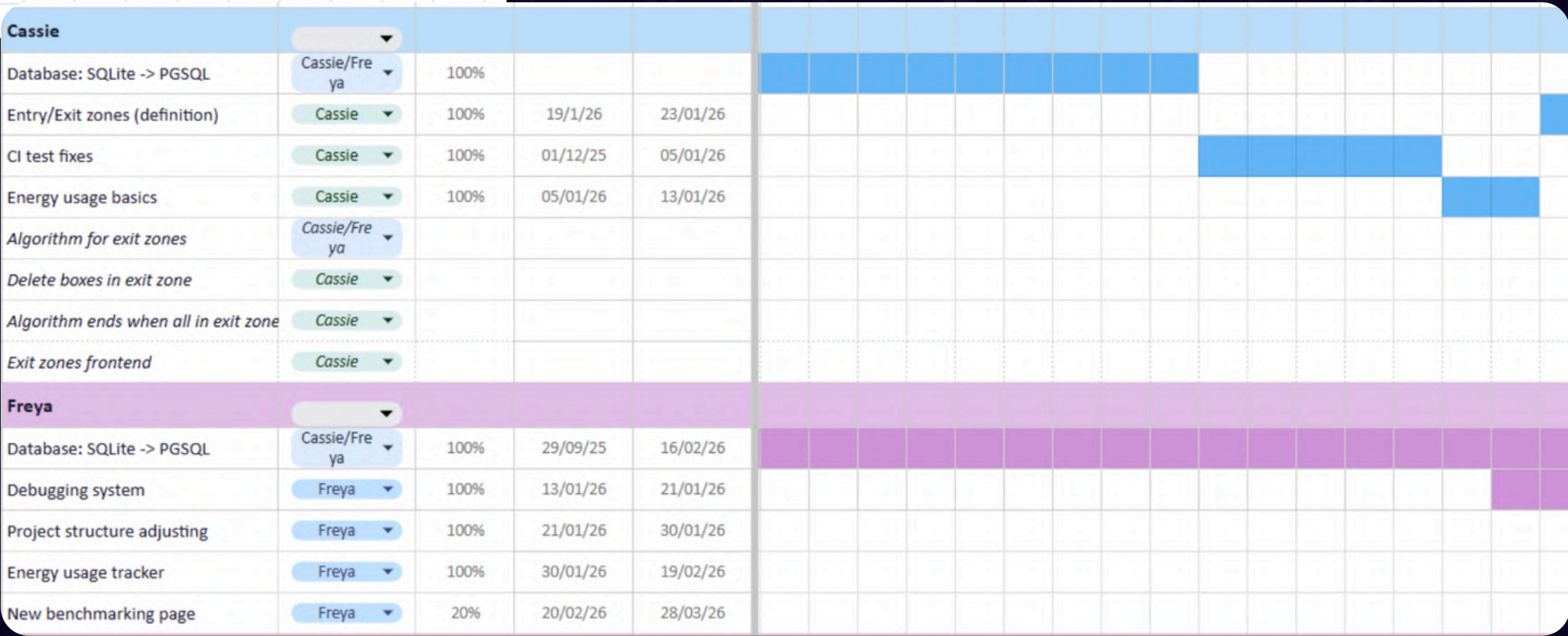
SOLUTIONS

- Put together a plan, started using Agile and doing retros
- Reached out more to Ashby (someone who worked on Robosim last year)
- More meetings and communication



GANTT CHART

- Helped with organisation and direction heading into TB2
- Kept everyone on the same page about plans



AGILE DEVELOPMENT

Rotating project manager

- Experience being responsible
- Active role in meetings

Bi-weekly sprints

- Ability to reflect and communicate
- Review what we need to adapt with retros!

Reflection from last sprints 'improvements':

- Have decent amount of achievable tasks every week

We have been assigning tasks and getting through them. Been adding them in 'Ready' for people to take on.

- Make sure we all get our agreed tasks done

Been communicating more which has been pushing us to get stuff done

- Update each other frequently

Met twice this week to prep which helped update eachother more clearly

- Clear with the plan

Has been clear but slightly over achievable

What went well?

- Feedback from Testing Day gave us ideas of improvements to make
- We have been really improving our repo helping us jump up the rankings
- More consistent commits
- Transparent boxes implemented
- System almost deployed only a couple more errors

What could have gone better?

- Probably set ourselves too many ambitious goals, will break them down for this next sprint to the beta
- Errors we've come across along the way has been what has set us back

Any team working problems?

- Not so much, meetings have been more efficient

How can we fix?

- Really want system deployed
- More frequent PRs
- Entrance/exit zones done
- Some improvements from Testing Day
- More features complete: perspective view and another on our list!

This sprints PM: Raymond

REPO IMPROVEMENTS

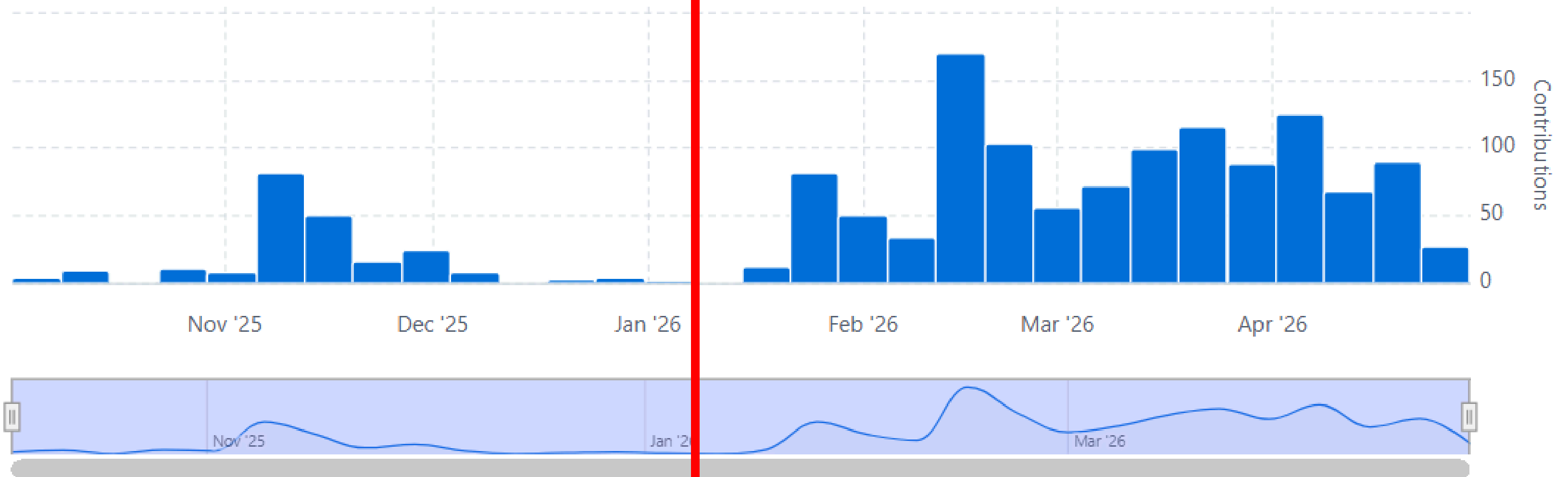
☐	 Energy usage tracker ✖ backend difficult enhancement frontend	6		8
#224 by freyadonnelly was merged on Feb 27 • Approved 5 of 12 tasks ➔ Beta Release				
☐	 Liveview feature ✖ difficult enhancement frontend	1		11
#220 by MincheolShinBristol was merged on Feb 23 • Approved 5 of 12 tasks ➔ Beta Release				
☐	 Finish CI/CD ✔ backend CI database DevOps difficult	3		15
#216 by ZiyangWang2615205 was merged on Feb 27 • Approved 4 of 12 tasks ➔ Beta Release				

- Giving more in depth feedback and code reviews on our PRs
- Breaking our tasks up on the Kanban so we can all see what we're working on
- In-review section on Kanban to encourage code reviews

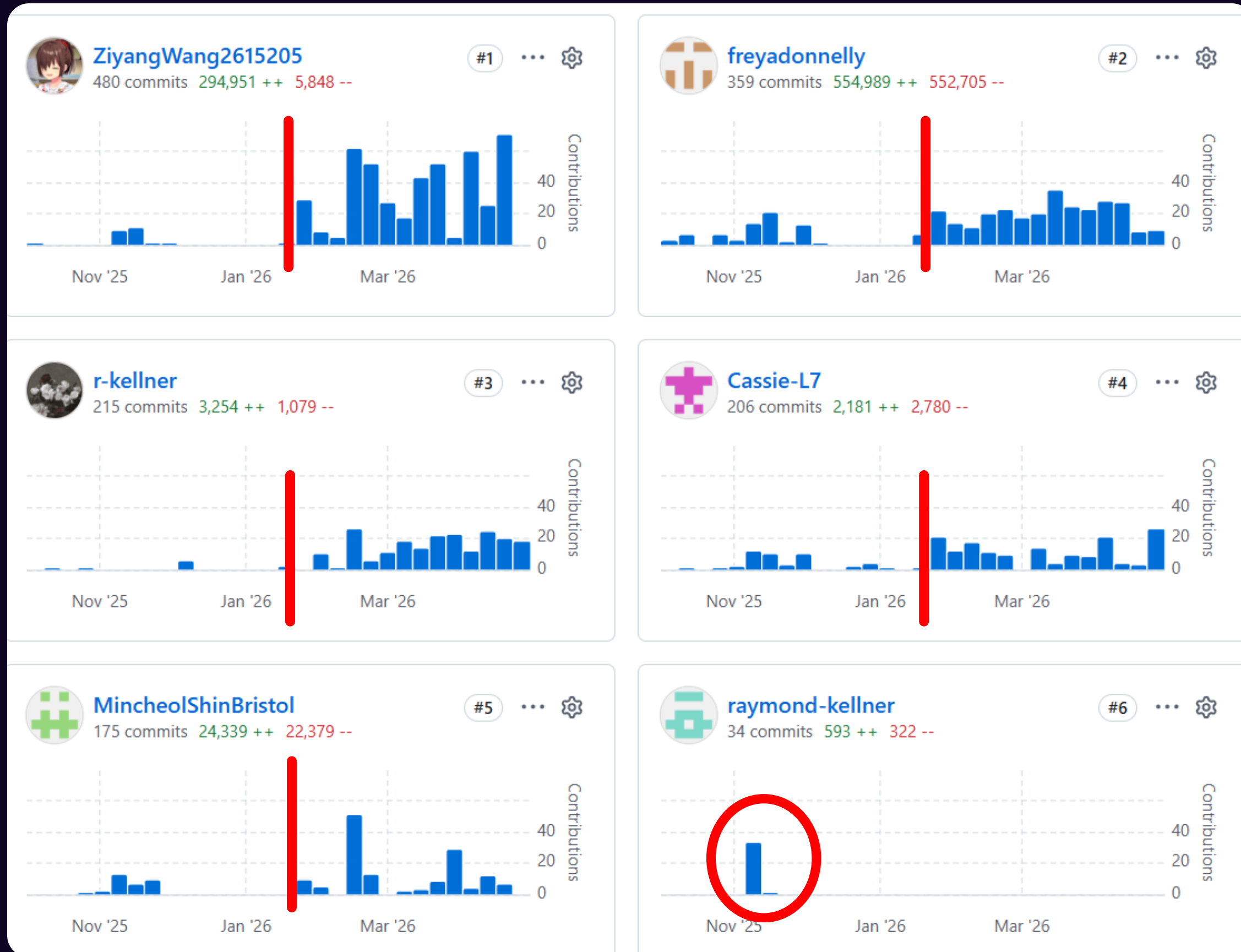
DID IT WORK?

BEFORE OUR CHANGES

AFTER OUR CHANGES!



DID IT WORK?



HANDOVER

LAST YEAR

- Instructions all over the place, >4 different documents & chain of links to follow
- Conflicting instructions on how to set up project!
- Only suggested improvement was replacing the entire db!

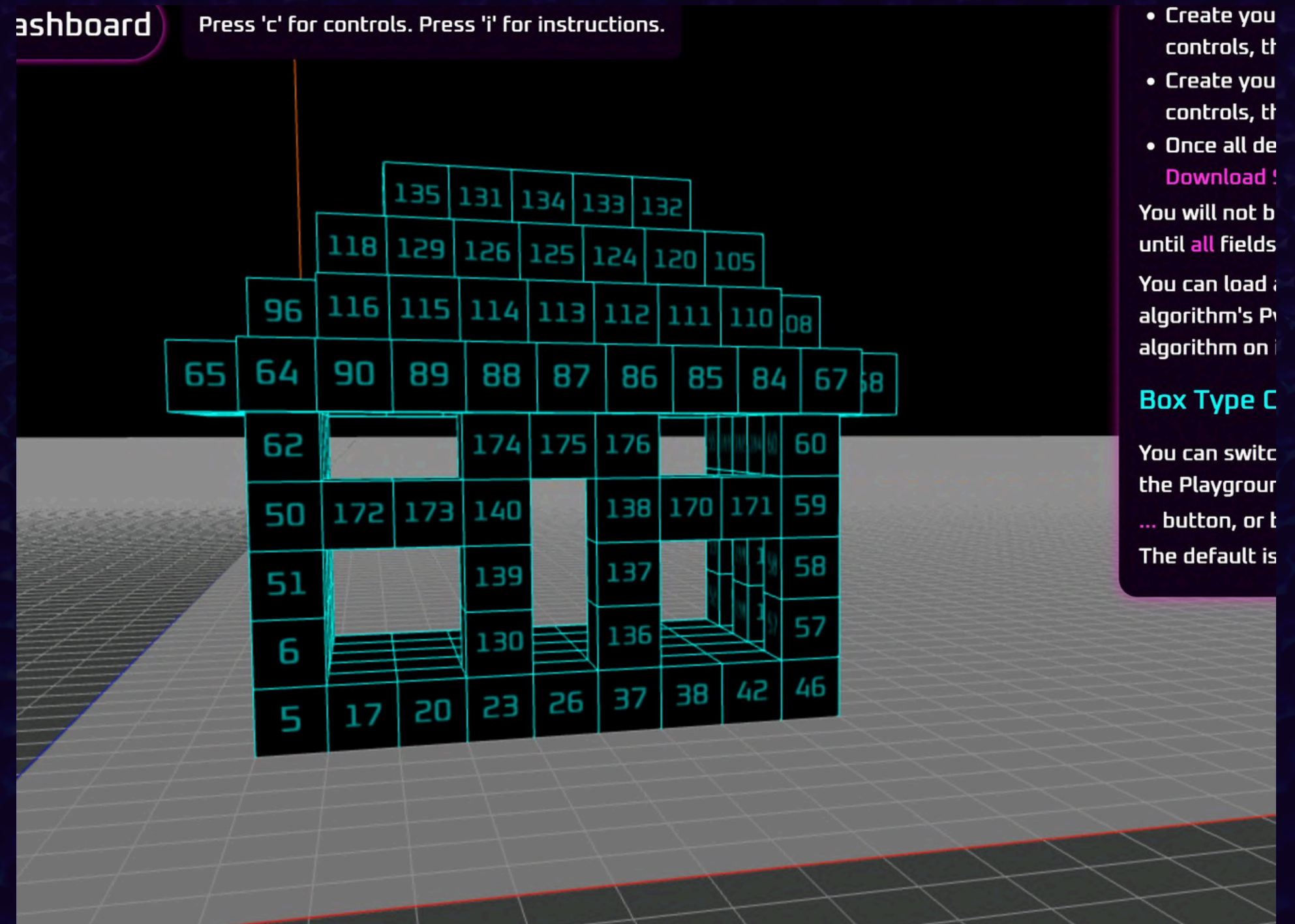


THIS YEAR!

- Central document with links to all key information e.g. advice on deployment
- Tested setup on another PC, deleted outdated files & wrote singular setup doc
- Several suggestions on where to start based off remaining ideas we had!

WHAT WE'D DO DIFFERENTLY

- Rotating roles earlier on
- Agile development from the start
- More organised PRs and Kanban board
- Sort our environment variables right from the start
- Jumping into coding sooner!



WHAT WE'VE LEARNED!



- Confidence to tackle projects that seem impossible at first
- Real-world project skills
- Consistently releasing code
- Lots more about unit tests
- Full-stack development